

BORON FAMILY

SECTION 1 — NCERT STATEMENT QUESTIONS (Q1-Q16)

These questions are based on the NCERT Group 13 chapter. Read each set and write the number of correct statements.

Q1. Occurrence, abundance and electronic configuration

1. Boron Family shows a gradual change from non-metallic to metallic character: boron is a typical non-metal, aluminium is metallic, and Ga, In and Tl are mainly metallic.
2. Boron occurs mainly in combined form; important occurrences include orthoboric acid, borax and kernite.
3. Borax occurs in India at Puga Valley in Ladakh and Sambhar Lake in Rajasthan.
4. Boron is a rare crustal element.
5. Natural boron consists mainly of ^{10}B and ^{11}B , and ^{11}B is the more abundant isotope.
6. Aluminium is the most abundant metal and the third most abundant element in Earth's crust after oxygen and silicon.
7. Bauxite and cryolite are important minerals of aluminium.
8. Aluminium is also mentioned as occurring as mica in Madhya Pradesh, Karnataka, Orissa and Jammu.
9. Gallium, indium and thallium are much less abundant than aluminium.
10. The valence-shell configuration of every Group 13 member is ns^2np^1 .
11. B and Al possess noble-gas cores; Ga and In have a noble-gas core plus d^{10} electrons.
12. Thallium has a noble-gas core together with filled f^{14} and d^{10} subshells.
13. The difference in inner electronic cores affects size, ionisation enthalpy and consequently the chemistry of the group.
14. Kernite and borax have exactly the same degree of hydration.

Number of correct statements: _____

Q2. Periodic anomalies and their causes

1. Atomic radius would normally increase down Group 13 because each successive element acquires an additional electron shell.
2. Gallium is smaller than aluminium because the intervening 3d electrons shield the increased nuclear charge poorly.
3. The fall in first ionisation enthalpy from B to Al is mainly associated with the increase in atomic size.
4. The small rise in first ionisation enthalpy from Al to Ga is related to poor shielding by the filled 3d electrons.
5. The rise in first ionisation enthalpy from In to Tl is related to poor shielding by intervening d and f electrons.
6. For every Group 13 element, the successive ionisation enthalpies obey $IE_1 < IE_2 < IE_3$.
7. The sum of the first three ionisation enthalpies is high, so direct formation of isolated M^{3+} ions is energetically costly.
8. Electronegativity falls sharply from B to Al and then shows only small irregular changes among the heavier members.
9. The irregular electronegativity behaviour is connected with the non-smooth variation of atomic size down the group.
10. Boron has an unusually high melting point because of its strong covalent crystalline lattice.
11. Gallium has an unusually low melting point but a high boiling point, giving it a wide liquid range.
12. The wide liquid range of gallium makes it useful for measuring high temperatures.
13. Density shows an overall increase from B to Tl despite other irregular physical-property trends.
14. The Ga anomaly demonstrates that Group 13 physical properties do not vary smoothly with atomic number.
15. Filled 3d electrons shield nuclear charge more effectively than s and p electrons and therefore make Ga larger than Al.

Number of correct statements: _____

Q3. MASTER PERIODIC-ORDER AUDIT — GROUP 13

1. Atomic/metallic radius: $\text{B} < \text{Ga} < \text{Al} < \text{In} < \text{Tl}$.
2. Trivalent ionic radius (tabulated M^{3+} values): $\text{B}^{3+} < \text{Al}^{3+} < \text{Ga}^{3+} < \text{In}^{3+} < \text{Tl}^{3+}$.
3. Monovalent ionic radius among the heavier ions: $\text{Ga}^+ < \text{In}^+ < \text{Tl}^+$.
4. First ionisation enthalpy: $\text{In} < \text{Al} < \text{Ga} < \text{Tl} < \text{B}$.
5. Second ionisation enthalpy: $\text{Al} < \text{In} < \text{Tl} < \text{Ga} < \text{B}$.
6. Third ionisation enthalpy: $\text{Al} < \text{In} < \text{Tl} < \text{Ga} < \text{B}$.
7. Correct third-ionisation-enthalpy order is $\text{In} < \text{Al} < \text{Tl} < \text{Ga} < \text{B}$.

8. Sum of the first three ionisation enthalpies: $\text{In} < \text{Al} < \text{Tl} < \text{Ga} < \text{B}$.
9. Electronegativity: $\text{Al} < \text{Ga} < \text{In} < \text{Tl} < \text{B}$.
10. Density: $\text{B} < \text{Al} < \text{Ga} < \text{In} < \text{Tl}$.
11. Melting point: $\text{Ga} < \text{Tl} < \text{In} < \text{Al} < \text{B}$.
12. Correct melting-point order is $\text{Ga} < \text{In} < \text{Tl} < \text{Al} < \text{B}$.
13. Boiling point: $\text{Tl} < \text{In} < \text{Ga} < \text{Al} < \text{B}$.
14. Standard reduction potential $E^\circ(M^{3+}/M)$: $\text{Al} < \text{Ga} < \text{In} < \text{Tl}$.
15. Stability of the +1 oxidation state: $\text{Al} < \text{Ga} < \text{In} < \text{Tl}$.
16. Metallic character generally increases in the order $\text{B} < \text{Al} < \text{Ga} < \text{In} < \text{Tl}$.
17. Acidic character of the normal oxides decreases overall from B toward Tl; basic character increases in the reverse sense.
18. Lewis acidity of boron trihalides: $\text{BF}_3 < \text{BCl}_3 < \text{BBr}_3 < \text{BI}_3$.
19. Halogen-to-boron $\text{p}\pi\text{-p}\pi$ donation: $\text{BI}_3 < \text{BBr}_3 < \text{BCl}_3 < \text{BF}_3$.
20. B-X bond length: $\text{BF}_3 < \text{BCl}_3 < \text{BBr}_3 < \text{BI}_3$.

Number of correct statements: _____

Q4. Chemical-trend logic

1. The usual group oxidation state is +3 because the valence configuration is ns^2np^1 .
2. Boron does not normally form B^{3+} because the sum of its first three ionisation enthalpies is very high; covalent bonding is therefore favoured.
3. Aluminium can form Al^{3+} much more readily than boron and is strongly electropositive.
4. Ga, In and Tl can show both +1 and +3 oxidation states.
5. The stability of the +1 oxidation state increases in the order $\text{Al} < \text{Ga} < \text{In} < \text{Tl}$.
6. Poor shielding by intervening d and f subshells causes the ns^2 pair to be held more strongly down the group.
7. Reduced participation of the ns^2 pair in bonding gives the textbook inert-pair trend.
8. Tl(I) is more stable than Tl(III) , so Tl(III) compounds are strongly oxidising.
9. Compounds in the +1 state are generally more ionic than related +3 compounds.
10. Monomeric trihalides are electron deficient and therefore behave as Lewis acids.
11. AlCl_3 can reduce electron deficiency by dimerising to halogen-bridged Al_2Cl_6 .
12. In the textbook model, boron cannot expand its octet; $[\text{BF}_4]^-$ is possible but $[\text{BF}_6]^{3-}$ is not.
13. B_2O_3 is acidic, Al_2O_3 and Ga_2O_3 are amphoteric, while basic character is greater for oxides of In and Tl.
14. In acidified aqueous solution aluminium forms octahedral $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$, whereas in alkali tetrahedral $[\text{Al}(\text{OH})_4]^-$ is important.
15. The inert-pair effect means the np^1 electron remains as an inert pair while the ns^2 electrons always form bonds.

Number of correct statements: _____

Q5. Borax: identity, structure, hydrolysis and heating

1. Borax is described as the most important compound of boron.
2. Borax is a white crystalline solid.
3. Its usual composition is written as $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$.
4. Its more structural representation is $\text{Na}_2[\text{B}_4\text{O}_5(\text{OH})_4] \cdot 8\text{H}_2\text{O}$.
5. The hydrated borax anion contains a tetranuclear boron-oxygen unit.
6. Borax dissolves in water to give an alkaline solution.
7. The hydrolysis may be represented as $\text{Na}_2\text{B}_4\text{O}_7 + 7\text{H}_2\text{O} \rightarrow 2\text{NaOH} + 4\text{H}_3\text{BO}_3$.
8. On heating, borax first loses water and swells.
9. Further heating produces a transparent liquid.
10. On cooling, this transparent liquid solidifies to a glass-like borax bead.
11. Strong heating ultimately gives sodium metaborate and B_2O_3 .
12. The overall heating sequence may be represented as $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O} \rightarrow \text{Na}_2\text{B}_4\text{O}_7 \rightarrow 2\text{NaBO}_2 + \text{B}_2\text{O}_3$.
13. Borax solution is acidic because it releases free H^+ ions directly.
14. The ten waters in $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$ are all present only as external lattice water.

Number of correct statements: _____

Q6. Borax bead test

1. Metaborates of many transition metals possess characteristic colours.

2. This colour formation is the basis of the borax bead test.
3. Borax is heated on a platinum-wire loop in a Bunsen flame during the classic test.
4. B_2O_3 formed in the bead can dissolve some metallic oxides.
5. With CoO , a blue bead is obtained.
6. The coloured cobalt product is cobalt metaborate, $Co(BO_2)_2$.
7. The reaction $CoO + B_2O_3 \rightarrow Co(BO_2)_2$ is consistent with formation of the blue bead.
8. The test is useful for identifying certain transition-metal ions.
9. Formation of the glassy bead follows dehydration and decomposition of borax.
10. The transparent bead acts as a solvent for certain metal oxides.
11. The cobalt bead is red because CoO is reduced to metallic cobalt.
12. Borax cannot be used in qualitative analysis because B_2O_3 is chemically inert toward metal oxides.
13. The colour arises from formation of a borate/metaborate species in the glass.
14. The bead test makes use of the glass-forming nature of boron oxide.

Number of correct statements: _____

Q7. Orthoboric acid: preparation, physical properties and structure

1. Orthoboric acid has the formula H_3BO_3 , commonly written structurally as $B(OH)_3$.
2. It is a white crystalline solid.
3. It has a soapy touch.
4. It is sparingly soluble in cold water.
5. It is much more soluble in hot water.
6. It can be prepared by acidifying an aqueous solution of borax.
7. The preparation may be represented as $Na_2B_4O_7 + 2HCl + 5H_2O \rightarrow 2NaCl + 4B(OH)_3$.
8. Hydrolysis of many boron halides can give boric acid.
9. Hydrolysis of boron hydrides can also give boric acid.
10. In the solid, planar BO_3 units are present.
11. Hydrogen bonding joins these units into a layered structure.
12. The solid structure is built only from tetrahedral BO_4 units.
13. Boric acid is described as a strong mineral acid.
14. Its solubility decreases sharply on heating water.

Number of correct statements: _____

Q8. Orthoboric acid: acidity, heating and mild-antiseptic use

1. Boric acid is a weak monobasic acid.
2. It is not a protonic acid in the usual sense.
3. It behaves as a Lewis acid by accepting OH^- from water.
4. The acid-base equilibrium may be represented as $B(OH)_3 + 2H_2O \rightleftharpoons [B(OH)_4]^- + H_3O^+$.
5. Its weakness is explained by the fact that it does not directly release H^+ on its own.
6. On heating above the stated temperature range, orthoboric acid forms metaboric acid, HBO_2 .
7. Further heating of metaboric acid gives boric oxide, B_2O_3 .
8. The thermal sequence may therefore be summarised as $H_3BO_3 \rightarrow HBO_2 \rightarrow B_2O_3$.
9. An aqueous solution of orthoboric acid is listed as a mild antiseptic.
10. Boric acid is tribasic because all three O-H hydrogens ionise independently in three normal dissociation steps.
11. Formation of $[B(OH)_4]^-$ is consistent with boron accepting an electron pair from hydroxide.
12. The final oxide B_2O_3 is the dehydration product of boric acid.
13. Metaboric acid is an intermediate in thermal dehydration.
14. Boric acid behaves as a strong reducing agent in water.

Number of correct statements: _____

Q9. Diborane: preparation, physical properties, combustion and hydrolysis

1. Diborane, B_2H_6 , is the simplest boron hydride highlighted in the chapter.
2. BF_3 can be reduced with $LiAlH_4$ in diethyl ether to prepare diborane.
3. The preparation is represented as $4BF_3 + 3LiAlH_4 \rightarrow 2B_2H_6 + 3LiF + 3AlF_3$.
4. A convenient laboratory route uses oxidation of $NaBH_4$ by iodine.
5. The laboratory equation is $2NaBH_4 + I_2 \rightarrow B_2H_6 + 2NaI + H_2$.
6. An industrial route uses BF_3 and sodium hydride at elevated temperature.

7. The industrial equation is $2BF_3 + 6NaH \rightarrow B_2H_6 + 6NaF$.
8. Diborane is a colourless, highly toxic gas.
9. It has a very low boiling point.
10. It catches fire spontaneously on exposure to air.
11. Combustion in oxygen gives B_2O_3 and H_2O and releases a large amount of energy.
12. Higher boranes are also described as spontaneously flammable in air.
13. Diborane is readily hydrolysed by water.
14. Complete hydrolysis gives boric acid and hydrogen.
15. $B_2O_3 + H_2O$ at room temperature is the standard direct preparation of diborane.
16. The combustion of diborane is strongly exothermic, with $\Delta_c H^\circ \approx -1976 \text{ kJ mol}^{-1}$.

Number of correct statements: _____

Q10. Diborane: Lewis-base cleavage, ammonia and borazine chemistry

1. Diborane undergoes cleavage reactions with Lewis bases.
2. The generic products of such cleavage may be written as $BH_3 \cdot L$ adducts.
3. $B_2H_6 + 2NMe_3 \rightarrow 2BH_3 \cdot NMe_3$ is an example of donor cleavage.
4. $B_2H_6 + 2CO \rightarrow 2BH_3 \cdot CO$ is another donor-adduct reaction.
5. Reaction of ammonia with diborane initially gives an adduct related to $B_2H_6 \cdot 2NH_3$.
6. The ammonia adduct is formulated as an ion pair containing $[BH_4]^-$.
7. Further heating of the ammonia-derived species gives borazine.
8. Borazine has formula $B_3N_3H_6$.
9. Borazine is called inorganic benzene because it has a six-membered ring containing alternating BH and NH groups.
10. The ammonia-to-borazine sequence is accompanied by release of hydrogen.
11. CO cannot act as a Lewis base toward a BH_3 fragment.
12. Borazine is a hydrocarbon composed only of carbon and hydrogen.
13. Ammonia converts diborane immediately and exclusively to BN without any intermediate.
14. The formation of $BH_3 \cdot L$ adducts reduces the electron deficiency of BH_3 fragments.

Number of correct statements: _____

Q11. Diborane: structure and bonding

1. The four terminal hydrogen atoms and the two boron atoms lie in one plane.
2. The two bridging hydrogen atoms lie above and below that plane.
3. The four terminal B-H bonds are normal two-centre two-electron bonds.
4. The two B-H-B bridge bonds are three-centre two-electron bonds.
5. The bridge bonds are often called banana bonds.
6. In the textbook bonding picture, each boron uses sp^3 hybrid orbitals.
7. One of the four sp^3 hybrids on each boron is shown without an electron in the electron-deficient description.
8. Diborane therefore contains two distinct types of B-H bonding.
9. In a 3c-2e bridge, two boron atoms and one hydrogen share two electrons.
10. All six hydrogen atoms in diborane are equivalent.
11. Diborane contains a conventional localised B=B double bond.
12. Every B-H bond in diborane is an ordinary localised 2c-2e bond.
13. The bridging description explains how diborane exists despite an insufficient number of electrons for only conventional bonds.
14. The structure is an example of electron-deficient multicentre bonding.

Number of correct statements: _____

Q12. Tetrahydridoborates / borohydrides

1. Boron forms a series of hydridoborates.
2. The most important ion discussed is $[BH_4]^-$.
3. $[BH_4]^-$ is tetrahedral.
4. Lithium and sodium tetrahydridoborates are also called borohydrides.
5. $LiBH_4$ and $NaBH_4$ may be prepared from the related metal hydrides and diborane in diethyl ether.

- The general equation is $2\text{MH} + \text{B}_2\text{H}_6 \rightarrow 2\text{M}^+[\text{BH}_4]^-$, where M = Li or Na.
- LiBH_4 and NaBH_4 are reducing agents.
- They are used in organic synthesis.
- They are useful starting materials for preparation of other metal borohydrides.
- The tetrahydridoborate ion contains four hydrogen atoms attached to boron.
- $[\text{BH}_4]^-$ is square planar.
- NaBH_4 is described as an oxidising agent because it accepts electrons.
- The borohydrides are prepared in the chapter by hydrolysis of diborane in water.
- The chemistry of $[\text{BH}_4]^-$ is consistent with hydride-donor reducing behaviour.

Number of correct statements: _____

NCERT USES AND APPLICATIONS (Q13-Q14)

Q13. Uses of boron, borax, boric acid and borides

- Boron is extremely hard and refractory.
- Boron combines high melting point with low density.
- Boron has very low electrical conductivity.
- Boron fibres are used in bullet-proof vests.
- Boron fibres are used in light composite materials for aircraft.
- ^{10}B has a high capacity for neutron absorption.
- Metal borides are therefore used in the nuclear industry as protective shields and control rods.
- Borax and boric acid are used in the manufacture of heat-resistant glass such as Pyrex.
- They are also used in making glass-wool and fibreglass.
- Borax is used as a flux in soldering metals.
- Borax is used in heat-, scratch- and stain-resistant glazed coatings on earthenware.
- Borax is a constituent of medicinal soaps.
- An aqueous solution of orthoboric acid is used as a mild antiseptic.
- Gallium, not borax, is the Group 13 material noted for high-temperature measurement because of its liquid range.

Number of correct statements: _____

Q14. Aluminium: properties, alloys and uses

- Aluminium is a bright silvery-white metal.
- It has high tensile strength.
- It has high electrical conductivity.
- It also has high thermal conductivity.
- On a weight-for-weight basis, its electrical conductivity is stated to be about twice that of copper.
- Aluminium forms alloys with Cu, Mn, Mg, Si and Zn.
- Aluminium and its alloys can be made into pipes and tubes.
- They can also be fabricated into rods and wires.
- Plates and foils are common fabricated forms.
- Aluminium is used in packaging.
- It is used in utensil making.
- It is used in construction.
- It is important in aeroplane and transportation industries.
- The chapter notes that domestic use of aluminium and its compounds has been reduced because of concern about their toxic nature.
- Aluminium is avoided in engineering because it cannot form useful alloys.
- Aluminium forms the light structural alloys duralumin (with Cu, Mn and Mg) and magnalium (with Mg).

Number of correct statements: _____

Q15. General reactions and reactivity

- Crystalline boron is comparatively unreactive under ordinary conditions.
- Aluminium is protected by a thin oxide film.
- On heating in air, amorphous boron gives B_2O_3 .
- On heating in air, aluminium gives Al_2O_3 .
- At high temperature, boron and aluminium form nitrides with dinitrogen.
- B_2O_3 reacts with basic metallic oxides to form borates.
- Boron does not react with ordinary acids and alkalis at moderate temperature.
- Aluminium dissolves in dilute HCl with release of hydrogen.
- Concentrated nitric acid makes aluminium passive.

- Aluminium dissolves in aqueous alkali to give tetrahydroaluminate(III) and hydrogen.
- Group 13 elements react with halogens to give trihalides, with TlI_3 noted as an exception.
- Trivalent chlorides are sufficiently covalent to undergo hydrolysis in water.
- White fumes around anhydrous AlCl_3 result from partial hydrolysis by atmospheric moisture and formation of moist HCl fumes.
- Concentrated nitric acid attacks aluminium rapidly because it removes the protective oxide layer.

Number of correct statements: _____

Q16. Assertion-reason logic

- Pair A: Ga is smaller than Al; poor shielding by 3d electrons is the reason.
- Pair B: First ionisation enthalpy falls from B to Al; increased atomic size is the main reason.
- Pair C: Electronegativity falls from B to Al and then rises slightly; irregular atomic size is the stated reason.
- Pair D: The +1 state becomes more stable toward Tl; tighter holding of the ns^2 pair due to poor d/f shielding is the reason.
- Pair E: Tl(III) is oxidising; Tl(I) is more stable than Tl(III) is the reason.
- Pair F: Ga is useful in high-temperature measurement; it combines a very low melting point with a high boiling point.
- Pair G: Boron has an unusually high melting point; a strong crystalline lattice is the reason.
- Pair H: Borax solution is alkaline; tetraborate hydrolysis produces basic species.
- Pair I: Boric acid is monobasic; it accepts OH^- from water rather than directly releasing three protons.
- Pair J: AlCl_3 fumes in moist air; partial hydrolysis liberates HCl that appears as moist fumes.
- Pair K: $[\text{BF}_6]^{3-}$ is not formed in the textbook model; boron cannot expand its octet beyond covalence four.
- Pair L: Diborane is electron deficient; its B-H-B bridges are described by 3c-2e bonding.
- Pair M: Aluminium is passive in concentrated HNO_3 ; a protective oxide film is maintained.
- Pair N: B_2O_3 is acidic; boron is the most metallic member of Group 13 is the reason.

Number of correct statements: _____

SECTION 2 — SINGLE CORRECT (Q17-Q31)

Each question has six statements. Choose the option that lists all correct statements and no wrong statement.

Q17.

Consider the following statements.

- NH_3 , primary amines and secondary amines are grouped as unsymmetrical-cleavage donors.
- Tertiary amines can give symmetrical neutral $\text{BH}_3 \cdot \text{L}$ adducts.
- Hydride ion can convert a BH_3 fragment into BH_4^- .
- CO can act as a donor toward an electron-deficient BH_3 fragment.
- Diborane cleavage requires no donor-acceptor interaction at boron.
- Every donor gives exactly the same diborane-cleavage product.

Which option gives the complete set of correct statements?

- 1, 2, 3, 5
- 1, 2, 3, 4
- 1, 2, 4, 6
- 1, 3, 4, 5

Q18.

Consider the following statements.

- The compound written TlI_3 is represented as Tl^+ and I_3^- in the classical description.
- The triiodide ion is linear.
- The central iodine of I_3^- is described as sp^3d hybridised in the elementary VSEPR model.
- All three iodine atoms in I_3^- have identical electronic environments.
- TlI_3 is correctly described as TlI_3^+ with three independent I^- ions.
- The inert-pair effect contributes to the relative stability of Tl(I) .

Which option gives the complete set of correct statements?

- 1) 1, 2, 3, 4
- 2) 1, 2, 5, 6
- 3) 1, 3, 4, 6
- 4) 1, 2, 3, 6

Q19.

Consider the following statements.

1. Borazine and benzene are isoelectronic in the usual comparison.
2. Both possess planar six-membered ring skeletons.
3. Borazine contains alternating boron and nitrogen atoms.
4. Polar B-N bonding makes borazine more likely to undergo addition and hydrolysis than benzene.
5. Borazine and benzene show identical reactivity toward HCl.
6. The charge distribution and bond polarity in borazine are identical to those in benzene.

Which option gives the complete set of correct statements?

- 1) 1, 2, 3, 4
- 2) 1, 2, 3, 5
- 3) 1, 2, 4, 6
- 4) 1, 3, 4, 5

Q20.

Consider the following statements.

1. BeO and Al_2O_3 are amphoteric.
2. Their chlorides show covalent character and association or bridging.
3. Be_2C and Al_4C_3 give CH_4 on hydrolysis.
4. Be_3N_2 and AlN give NH_3 on hydrolysis.
5. Both elements form strongly basic oxides only.
6. Their aqueous ions are unaffected by hydration or hydrolysis.

Which option gives the complete set of correct statements?

- 1) 1, 2, 3, 5
- 2) 1, 2, 4, 6
- 3) 1, 2, 3, 4
- 4) 1, 3, 4, 5

Q21.

Consider the following statements.

1. Hydrated borax can be written $\text{Na}_2[\text{B}_4\text{O}_5(\text{OH})_4] \cdot 8\text{H}_2\text{O}$.
2. The tetraborate unit contains two trigonal and two tetrahedral boron centres.
3. The conventional tetraborate framework contains five B-O-B linkages.
4. The common decahydrate notation is equivalent to four structural OH groups plus eight lattice waters.
5. Each of the four boron centres carries one OH group in the standard drawing.
6. The localised model contains four terminal B=O double bonds.

Which option gives the complete set of correct statements?

- 1) 1, 2, 3, 4, 5
- 2) 1, 2, 3, 4, 6
- 3) 1, 2, 3, 5, 6
- 4) 1, 2, 4, 5, 6

Q22.

Consider the following statements.

1. Reduction of BF_3 with LiAlH_4 in dry ether can give diborane.
2. Diborane is a toxic, electron-deficient gas.
3. Complete hydrolysis gives boric acid and hydrogen.
4. Diborane behaves as a Lewis acid toward suitable donors.
5. Its bonding is completely identical to the ordinary ethane model.
6. It contains four terminal B-H bonds and two B-H-B bridges.

Which option gives the complete set of correct statements?

- 1) 1, 2, 3, 4, 5
- 2) 1, 2, 3, 5, 6
- 3) 1, 2, 4, 5, 6
- 4) 1, 2, 3, 4, 6

Q23.

Consider the following statements.

1. Diborane can accept electron-pair donors through BH_3 fragments.

2. BF_3 acts as a Lewis acid despite fluorine-to-boron back donation.
3. AlCl_3 is a Lewis acid and commonly forms donor adducts.
4. BF_4^- is the most electron-deficient Lewis acid in the set.
5. BeF_2 can accept additional fluoride to form a fluoro complex.
6. The familiar Lewis-acidity order is $\text{BF}_3 < \text{BCl}_3 < \text{BBr}_3 < \text{BI}_3$.

Which option gives the complete set of correct statements?

- 1) 1, 2, 3, 5, 6
- 2) 1, 2, 3, 4, 6
- 3) 1, 2, 4, 5, 6
- 4) 1, 3, 4, 5, 6

Q24.

Consider the reaction given below and then read the statements.



1. At elevated temperature the boron-containing product P is diborane.
2. The sodium salt Q is NaF.
3. P is electron deficient.
4. P can form donor adducts with suitable phosphorus or nitrogen donors.
5. Acidified MnO_2 readily oxidises Q to elemental fluorine under ordinary laboratory conditions.
6. Q gives a white CaF_2 precipitate with Ca^{2+} solution.

Which option gives the complete set of correct statements?

- 1) 1, 2, 3, 4, 5
- 2) 1, 2, 3, 5, 6
- 3) 1, 2, 3, 4, 6
- 4) 1, 2, 4, 5, 6

Q25.

Consider the following statements about boric acid, H_3BO_3 .

1. In one H_3BO_3 molecule, boron is bonded to three -OH groups and the BO_3 part is planar.
2. In water, boric acid behaves as a weak monobasic Lewis acid.
3. It accepts OH^- from water to form $[\text{B}(\text{OH})_4]^-$ instead of directly giving three H^+ ions.
4. Hydrogen bonding is important in solid boric acid.
5. Boric acid is a strong tribasic acid because all three O-H hydrogens ionise separately.
6. Polyhydroxy compounds such as glycerol or mannitol can form complexes with boric acid and increase its apparent acidity.

Which option gives the complete set of correct statements?

- 1) 1, 2, 3, 4, 5
- 2) 1, 2, 3, 4, 6
- 3) 1, 2, 3, 5, 6
- 4) 1, 2, 4, 5, 6

Q26.

Consider the following statements.

1. An aqueous borax solution is alkaline because of hydrolysis.
2. $\text{B}(\text{OH})_3$ and $[\text{B}(\text{OH})_4]^-$ participate in the aqueous equilibrium description.
3. Borax behaves as a strong mineral acid in water.
4. Acidification of borax can precipitate boric acid.
5. Borax-boric-acid mixtures are useful in buffer and titration chemistry.
6. B_2O_3 is the only stable boron species in dilute aqueous borax.

Which option gives the complete set of correct statements?

- 1) 1, 2, 3, 4
- 2) 1, 2, 5, 6
- 3) 1, 3, 4, 5
- 4) 1, 2, 4, 5

Q27.

Consider the following statements.

1. The +1 oxidation state becomes increasingly stable down Group 13.
2. The inert-pair effect helps stabilise Tl(I) .
3. Tl(III) salts can behave as oxidising agents.
4. TlI_3 is commonly interpreted as Tl(I) triiodide.
5. Tl(I) is less stable than Tl(III) in aqueous chemistry.
6. No compound containing thallium in the +3 oxidation state can exist.

Which option gives the complete set of correct statements?

- 1) 1, 2, 3, 4

- 2) 1, 2, 3, 5
- 3) 1, 2, 4, 6
- 4) 1, 3, 4, 5

Q28.

Consider the following statements.

1. Washing-soda-containing alkaline material can attack aluminium.
2. Strong alkali can form soluble aluminate or hydroxoaluminate while evolving H_2 .
3. Concentrated HNO_3 can passivate aluminium.
4. Aluminium is used in transmission conductors because of conductivity and low density.
5. $Al(OH)_3$ is used as an antacid.
6. Keeping water overnight in bare aluminium utensils is universally recommended in the supplied practical notes.

Which option gives the complete set of correct statements?

- 1) 1, 2, 3, 4, 6
- 2) 1, 2, 3, 5, 6
- 3) 1, 2, 3, 4, 5
- 4) 1, 2, 4, 5, 6

Q29.

Consider the following statements.

1. Anhydrous $AlCl_3$ is mainly covalent.
2. Associated Al_2Cl_6 species occur in the vapour under suitable conditions.
3. Solid anhydrous $AlCl_3$ is described as polymeric or high-coordinate.
4. Moisture causes hydrolysis and visible HCl fumes.
5. In water, strongly hydrated aluminium species such as $[Al(H_2O)_6]^{3+}$ are stabilised.
6. The hydration enthalpy of Al^{3+} is too small to influence aqueous forms present.

Which option gives the complete set of correct statements?

- 1) 1, 2, 3, 4, 6
- 2) 1, 2, 3, 4, 5
- 3) 1, 2, 3, 5, 6
- 4) 1, 2, 4, 5, 6

Q30.

Consider the following statements.

1. Near 373 K, H_3BO_3 first gives metaboric acid, HBO_2 .
2. Further heating near 413 K gives tetraboric acid, $H_2B_4O_7$.
3. Red heat ultimately gives B_2O_3 .
4. Borax is the necessary first dehydration product of pure boric acid.
5. B_2O_3 can be rehydrated to boric acid under suitable conditions.
6. The sequence proves that boric acid is a strong tribasic proton donor.

Which option gives the complete set of correct statements?

- 1) 1, 2, 3, 4
- 2) 1, 2, 5, 6
- 3) 1, 3, 4, 5
- 4) 1, 2, 3, 5

Q31.

Consider the following statements.

1. AlN gives NH_3 on hydrolysis.
2. Al_4C_3 gives CH_4 on hydrolysis.
3. Al_4C_3 is classified as a methanide carbide in this reaction context.
4. Hydrolysis of AlN converts nitride nitrogen into ammonia.
5. Al_4C_3 gives acetylene as its characteristic hydrolysis gas.
6. The gaseous products from AlN and Al_4C_3 are respectively NH_3 and CH_4 .

Which option gives the complete set of correct statements?

- 1) 1, 2, 3, 4, 5
- 2) 1, 2, 3, 5, 6
- 3) 1, 2, 3, 4, 6
- 4) 1, 2, 4, 5, 6

SECTION 3 — ONE OR MORE CORRECT (Q32-Q37)

One or more statements may be correct. Select all correct statements.

Q32.

Select all correct statements.

- 1) Bridge B-H contacts are longer and weaker than terminal B-H bonds.
- 2) Terminal B-H bonds are ordinary 2c-2e bonds.
- 3) Each B-H-B bridge is a 3c-2e bond.
- 4) The two B atoms and four terminal H atoms form the maximum coplanar set.
- 5) All six hydrogen atoms are equivalent ordinary terminal hydrogens.
- 6) Boron is described as sp^2 hybridised in the classical bridge-bond model.

Q33.

Select all correct statements.

- 1) The joined species is represented as $Al_2(CH_3)_6$.
- 2) Two methyl groups are bridging.
- 3) Four methyl groups are terminal.
- 4) Bridging Al-C contacts are longer than terminal Al-C bonds.
- 5) The bridges are electron-deficient multicentre interactions.
- 6) The compound is correctly described as separated Al^{3+} and CH_3^- ions.

Q34.

Select all correct statements.

- 1) $Al(OH)_3$ is obtained as a white gelatinous precipitate.
- 2) It is insoluble in excess NH_4OH .
- 3) It dissolves in strong $NaOH$ to form aluminate or hydroxoaluminate species.
- 4) It dissolves in acids.
- 5) It is used as an antacid.
- 6) It is a strongly basic, non-amphoteric hydroxide.

Q35.

Select all correct statements.

- 1) $BaCl_2$ can precipitate sulfate as $BaSO_4$.
- 2) Na_2CO_3 can lead to a chromium(III) hydroxide-type precipitate through hydrolysis.
- 3) $(NH_4)_2S$ can produce a chromium(III) hydroxide-type precipitate in water.
- 4) A controlled amount of KOH can precipitate $Cr(OH)_3$.
- 5) A large excess of strong alkali can dissolve amphoteric $Cr(OH)_3$.
- 6) $BaCl_2$ precipitates chromium exclusively as barium chromate from untreated $Cr(III)$ alum.

Q36.

Select all correct statements.

- 1) In the stated flow, A is sodium metaborate, $NaBO_2$.
- 2) B represents sodium tetraborate or borax chemistry.
- 3) $Ca_2B_6O_{11} + 2Na_2CO_3 \rightarrow 2CaCO_3 + 2A + B$ is balanced for these identities.
- 4) The tetraborate framework of B contains five B-O-B linkages.
- 5) Strong heating of borax can regenerate $NaBO_2$ together with B_2O_3 .
- 6) Solid sodium metaborate must contain only isolated monomeric BO_2^- ions.

Q37.

Select all correct statements.

- 1) Salicylic acid can form a four-coordinate chelated boron complex.
- 2) The salicylate product is treated as optically resolvable in the supplied problem set.
- 3) Polyhydroxy ligands can enhance the apparent acidity of boric acid.
- 4) Methanol with concentrated H_2SO_4 can form a volatile borate ester that burns with a green-edged flame.
- 5) No reaction can occur between boric acid and salicylic acid because both are acids.
- 6) Chelation forces boron to remain strictly three-coordinate and trigonal.

SECTION 4 — MATCH THE COLUMNS (Q38-Q47)

Match each item in Column I with the chemically appropriate item(s) in Column II.

Q38.

Match Column I with Column II. One or more matches may apply where chemically appropriate.

Column I	Column II
A. W in $\text{B}_2\text{H}_6 + \text{H}_2\text{O} \rightarrow \text{W} + \text{H}_2$	P. B_2O_3
B. X after first heating of W	Q. HBO_2
C. Y after further condensation	R. $\text{H}_2\text{B}_4\text{O}_7$
D. Z after strongest heating	S. H_3BO_3

Your match: _____

Q39.

Match Column I with Column II. One or more matches may apply where chemically appropriate.

Column I	Column II
A. $\text{BBr}_3 + \text{H}_2$ (high temperature) $\rightarrow \text{B}$	P. borax-bead test
B. Borax + Cu salt $\rightarrow$ coloured metaborate bead	Q. reduction
C. AlCl_3 + atmospheric $\text{H}_2\text{O} \rightarrow \text{HCl}$ fumes	R. white fumes
D. $\text{Cr}_2\text{O}_3 + \text{Al} \rightarrow \text{Cr}$	S. hydrolysis

Your match: _____

Q40.

Match Column I with Column II. One or more matches may apply where chemically appropriate.

Column I	Column II
A. BF_3	P. trigonal-planar electron-deficient monomer
B. solid AlCl_3	Q. polymeric/high-coordinate environment
C. Al_2Cl_6 vapour dimer	R. two chloride bridges
D. B_2H_6	S. 3c-2e B-H-B bridges

Your match: _____

Q41.

Match Column I with Column II. One or more matches may apply where chemically appropriate.

Column I	Column II
A. NH_3 / MeNH_2 / Me_2NH	P. unsymmetrical cleavage in the question-bank classification
B. Me_3N	Q. symmetric neutral BH_3 -L adduct formation
C. CO	R. formation of BH_4^- salt
D. NaH	S. Lewis-base interaction with electron-deficient boron

Your match: _____

Q42.

Match Column I with Column II. One or more matches may apply where chemically appropriate.

Column I	Column II
A. Borazine	P. inorganic benzene
B. hexagonal BN	Q. inorganic-graphite-type layered solid
C. B_4C	R. very hard abrasive/boron-rich carbide
D. Al_4C_3	S. methanide carbide giving CH_4 on hydrolysis

Your match: _____

Q43.

Match List I with every applicable property in List II.

Column I	Column II
A. Borazine (borazole)	P. Lewis acid
B. Diborane	Q. Dimer
C. Aluminium chloride	R. Delocalisation of electrons
D. Boron trichloride	S. Hydrolyses in water

Your match: _____

Q44.

Set I. Match each reaction with its main product.

Column I	Column II
A. Borax, $\Delta \rightarrow$	P. SiO_2
B. $\text{B}_2\text{H}_6 + \text{H}_2\text{O} \rightarrow$	Q. H_2SiO_3
C. $\text{B}_2\text{H}_6 + \text{excess NH}_3, \Delta \rightarrow$	R. BN
D. SiCl_4 ; (i) H_2O , (ii) $1000^\circ\text{C} \rightarrow$	S. $\text{NaBO}_2 + \text{B}_2\text{O}_3$
	T. H_3BO_3

Set II. Match each reaction with its main product.

Column I	Column II
A. Borax, $\Delta \rightarrow$	P. BN
B. $\text{B}_2\text{H}_6 + \text{H}_2\text{O} \rightarrow$	Q. B_2H_6
C. $\text{B}_2\text{H}_6 + \text{excess NH}_3, \Delta \rightarrow$	R. H_3BO_3
D. $\text{BCl}_3 + \text{LiAlH}_4 \rightarrow$	S. $\text{NaBO}_2 + \text{B}_2\text{O}_3$

Your match: _____

Q45.

Set I. Match each reaction with every supported product characteristic.

Column I	Column II
A. Borazine + $\text{H}_2\text{O} \rightarrow$	P. One product is a gas
B. Borax + NH_4Cl , $400^\circ\text{K} \rightarrow$	Q. One product is a monobasic acid
C. $\text{BF}_3 + \text{NaH}$, $180^\circ\text{C} \rightarrow$	R. One product has a graphite-like structure
D. $\text{B}_2\text{H}_6 + \text{H}_2\text{O} \rightarrow$	S. One product turns mercurous-nitrate paper black or is a washing-powder brightener
	T. One product is an acidic oxide

Set II. Match each borax transformation with the indicated product set.

Column I	Column II
A. Borax, $\Delta \rightarrow$	P. $\text{Na}[\text{B}(\text{OH})_4] + \text{H}_3\text{BO}_3$
B. Borax + NH_4Cl , $\Delta \rightarrow$	Q. $\text{NaBO}_2 + \text{B}_2\text{O}_3$
C. Borax + $\text{HCl} \rightarrow$	R. BN
D. Borax + $\text{H}_2\text{O} \rightarrow$	S. H_3BO_3

Your match: _____

Q46.

Match each species with every applicable characteristic.

Column I	Column II
A. B_2H_6	P. Solid at room temperature
B. H_3BO_3	Q. Gas at room temperature
C. BF_3	R. On hydrolysis gives an oxyacid as one product
D. BI_3	S. Boron is sp^2 -hybridised
	T. Covalent compound

Your match: _____

Q47.

Set I. Match each substance with every applicable description.

Column I	Column II
A. Borax	P. Gives yellow precipitate with sodium cobaltinitrite
B. Potash alum	Q. sp ² -hybridised boron
C. Borazon	R. Structure similar to graphite
D. Borazole	S. sp ³ -hybridised boron

Set II. Match each substance with its common use or name.

Column I	Column II
A. Anhydrous AlCl ₃	P. Antiseptic
B. Alum	Q. Alloy
C. Ultramarine	R. Friedel-Crafts catalyst
D. Borax	S. Complex blue silicate
E. Magnalium	T. Tincal

Your match: _____

SECTION 5 — COUNT THE CORRECT STATEMENTS (Q48-Q54)

Enter the number of correct statements.

Q48.

How many statements are correct?

1. Borax is represented structurally as Na₂[B₄O₅(OH)₄] · 8H₂O.
2. Two boron atoms are trigonal and two are tetrahedral in the tetraborate unit.
3. The tetraborate framework contains five B-O-B linkages.
4. The localised problem-bank description assigns zero B=O π bonds to hydrated borax.
5. Strong heating produces NaBO₂ + B₂O₃.
6. The glassy bead can dissolve metal oxides and form coloured metaborates.
7. CoO gives a characteristic blue bead.
8. Borax is used as a primary standard in acidimetry.
9. Its aqueous solution is alkaline because of hydrolysis.
10. All four boron centres are tetrahedral.
11. Borax contains a conventional B=B π bond.
12. Borax can be converted through boric acid and B₂O₃ to elemental boron.

Number of correct statements: _____

Q49.

How many statements are correct?

1. B₂H₆ has 12 valence electrons.
2. Four terminal B-H bonds are normal 2c-2e bonds.
3. Two bridge B-H-B bonds are 3c-2e bonds.
4. Bridge B-H distances are longer than terminal B-H distances.
5. Bridge bonds are weaker than terminal B-H bonds.
6. The two boron atoms and four terminal hydrogens form the maximum coplanar set.
7. Bridge hydrogens lie above and below that plane.
8. Boron is described as sp³ hybridised in the classical model.
9. Tertiary amines can give symmetric BH₃ · L cleavage products.
10. NH₃, MeNH₂ and Me₂NH are grouped as unsymmetrical-cleavage donors in the problem set.
11. Every B-H contact is an ordinary 2c-2e bond.
12. The bridge B-H bonds are shorter and stronger than the terminal bonds.

Number of correct statements: _____

Q50.

How many statements are correct?

1. Anhydrous AlCl₃ is covalent.
2. In moist air it partially hydrolyses and gives HCl fumes.
3. Large hydration enthalpy helps stabilise ionic aluminium species in water.
4. [Al(H₂O)₆]³⁺ is octahedral.
5. [Al(OH)₄]⁻ is tetrahedral.
6. AlCl₃ can form the associated dimer Al₂Cl₆ in vapour.
7. Solid anhydrous AlCl₃ is described as polymeric/high-coordinate in the question bank.
8. AlCl₃ · 6H₂O should be treated as the same electron-deficient species as dry monomeric AlCl₃.
9. Chloride bridges reduce electron deficiency at aluminium.
10. Aluminium can show coordination numbers greater than four.
11. Hydrated Al³⁺ is destabilised by hydration, so water forces AlCl₃ to remain molecular.

12. Anhydrous and hydrated forms therefore need not have the same bonding description.

Number of correct statements: _____

Q51.

How many statements are correct?

1. Borazine has formula B₃N₃H₆.
2. It is called inorganic benzene.
3. The ring contains alternating B and N atoms.
4. B and N are sp² in the usual planar ring description.
5. B-N π interaction/back donation is included in the classical description.
6. Borazine is more reactive toward addition/hydrolysis than benzene.
7. Hydrolysis can give boric acid and ammonia.
8. Hydrogen may also appear in the source reaction representation.
9. High-temperature ammonia/boron-hydride chemistry can give BN.
10. Hexagonal BN has a graphite-like layered structure.
11. Borazine and benzene have exactly identical reactivity toward HCl.
12. Every B-N bond is non-polar because B and N have identical electronegativity.

Number of correct statements: _____

Q52.

How many statements are correct?

1. Boric acid is a weak monobasic Lewis acid.
2. It accepts OH⁻ from water rather than directly releasing three protons.
3. Solid boric acid contains hydrogen bonding.
4. Heating gives HBO₂ and ultimately B₂O₃.
5. Boric acid can be formed by hydrolysis of BCl₃.
6. Boric acid can be formed by hydrolysis of B₂H₆.
7. Acidification of borax gives boric acid.
8. Polyhydroxy compounds can enhance its apparent acidity by complex formation.
9. Salicylic acid can give a four-coordinate chelated boron product in the problem set.
10. Boric acid is a strong tribasic protonic acid.
11. Borate + methanol + concentrated H₂SO₄ can produce a volatile ester giving a green-edged flame.
12. No reaction is possible between boric acid and salicylic acid because both are acids.

Number of correct statements: _____

Q53.

How many statements are correct?

1. Borax can be acidified to boric acid and then dehydrated to B₂O₃.
2. B₂O₃ can be reduced by an active metal such as Mg to obtain boron.
3. Crude boron from active-metal reduction may require acid washing to remove impurities.
4. Boron can dissolve under suitable conditions in hot concentrated H₂SO₄.
5. Concentrated HNO₃ passivates aluminium.
6. Aluminium is used for transmission conductors because of its conductivity and low density.
7. Al + NaOH mixtures can evolve H₂ and are used in some drain-cleaning formulations.
8. Washing-soda-containing alkaline material can attack aluminium and form soluble aluminate species.
9. Al(OH)₃ can be used as an antacid.
10. AlN hydrolysis gives NH₃.
11. Al₄C₃ hydrolysis gives CH₄.
12. Calcium is the preferred clean reducing metal in the cited B₂O₃-to-boron isolation scheme.

Number of correct statements: _____

Q54.

How many statements are correct?

1. Ethyl borate, B(OC₂H₅)₃, burns with a green-edged flame.
2. Boron is obtained by pyrolysis of BI₃.
3. One mole of borax reacts with one mole of a monobasic acid.
4. In sodium peroxoborate, each boron atom is sp³-hybridised.
5. H₃BO₃ does not dissolve in aqueous HF.

6. $\text{B}_2\text{H}_6 + \text{excess NH}_3$ above 200°C gives borazine as the final product.

Numerical response: _____

SECTION 6 — NUMERICAL RESPONSE (20 QUESTIONS: Q55-Q74)

Each question has one numerical answer and no options. Write only the numerical value unless a unit is asked for.

Q55.

For one B_2H_6 molecule, what is the maximum number of atoms that can lie in a single plane in its standard structure?

Numerical response: _____

Q56.

How many terminal B–H bonds in one B_2H_6 molecule are ordinary two-centre two-electron (2c–2e) bonds?

Numerical response: _____

Q57.

How many B–H–B three-centre two-electron (3c–2e) bridge bonds are present in one B_2H_6 molecule?

Numerical response: _____

Q58.

Find the total number of valence electrons present in one B_2H_6 molecule.

Numerical response: _____

Q59.

In the tetraborate anion $[\text{B}_4\text{O}_5(\text{OH})_4]^{2-}$ of hydrated borax, how many boron atoms are trigonal (sp^3)?

Numerical response: _____

Q60.

In the tetraborate anion $[\text{B}_4\text{O}_5(\text{OH})_4]^{2-}$, how many boron atoms are tetrahedral (sp^3)?

Numerical response: _____

Q61.

In the standard structural representation of $[\text{B}_4\text{O}_5(\text{OH})_4]^{2-}$, how many B–O–B bridging oxygen atoms are present?

Numerical response: _____

Q62.

Using the standard $[\text{B}_4\text{O}_5(\text{OH})_4]^{2-}$ structure, count the total number of B–O σ bonds, including B–O(H) bonds.

Numerical response: _____

Q63.

Borax is written structurally as $\text{Na}_2[\text{B}_4\text{O}_5(\text{OH})_4] \cdot 8\text{H}_2\text{O}$. How many water molecules are present as lattice water per formula unit?

Numerical response: _____

Q64.

In one molecule of borazine, $\text{B}_3\text{N}_3\text{H}_6$, how many ring atoms are sp^2 -hybridised?

Numerical response: _____

Q65.

Assuming the standard planar structure of $\text{B}_3\text{N}_3\text{H}_6$, what is the maximum number of atoms of one borazine molecule that lie in the ring plane?

Numerical response: _____

Q66.

Complete hydrolysis of 0.25 mol B_2H_6 occurs according to $\text{B}_2\text{H}_6 + 6\text{H}_2\text{O} \rightarrow 2\text{H}_3\text{BO}_3 + 6\text{H}_2$. Find the moles of H_2 produced.

Numerical response: _____

Q67.

$\text{B}_2\text{H}_6 + 6\text{CH}_3\text{OH} \rightarrow 2\text{B}(\text{OCH}_3)_3 + 6\text{H}_2$. If 3.0 mol B_2H_6 reacts completely with excess methanol, how many moles of $\text{B}(\text{OCH}_3)_3$ are formed?

Numerical response: _____

Q68.

$\text{Na}_2\text{B}_4\text{O}_7 + 7\text{H}_2\text{O} \rightarrow 2\text{NaOH} + 4\text{H}_3\text{BO}_3$. If 0.50 mol $\text{Na}_2\text{B}_4\text{O}_7$ is completely hydrolysed, how many moles of H_3BO_3 are formed?

Numerical response: _____

Q69.

On strong heating, $2\text{H}_3\text{BO}_3 \rightarrow \text{B}_2\text{O}_3 + 3\text{H}_2\text{O}$. Starting with 4.0 mol H_3BO_3 , how many moles of B_2O_3 are obtained on complete dehydration?

Numerical response: _____

Q70.

$2\text{Al} + 2\text{NaOH} + 6\text{H}_2\text{O} \rightarrow 2\text{Na}[\text{Al}(\text{OH})_4] + 3\text{H}_2$. If 2.0 mol Al reacts completely with excess NaOH and water, how many moles of H_2 are evolved?

Numerical response: _____

Q71.

$2\text{NaBH}_4 + \text{I}_2 \rightarrow 2\text{NaI} + \text{B}_2\text{H}_6 + \text{H}_2$. If 6.0 mol NaBH_4 reacts completely with iodine, find the total moles of gaseous products formed.

Numerical response: _____

Q72.

How many of the following statements are correct? (i) Lewis acidity: $\text{BF}_3 < \text{BCl}_3 < \text{BBr}_3 < \text{BI}_3$; (ii) $\text{p}\pi\text{--p}\pi$ donation: $\text{BF}_3 > \text{BCl}_3 > \text{BBr}_3 > \text{BI}_3$; (iii) B–X bond length: $\text{BF}_3 < \text{BCl}_3 < \text{BBr}_3 < \text{BI}_3$; (iv) free BX_3 monomers are trigonal planar; (v) B–F has the strongest $\text{p}\pi\text{--p}\pi$ interaction; (vi) BI_3 has the strongest $\text{p}\pi\text{--p}\pi$ overlap; (vii) $[\text{BF}_4]^-$ is tetrahedral; (viii) BF_3 is a stronger Lewis acid than BI_3 .

Numerical response: _____

Q73.

Beryl is represented as $\text{Be}_n\text{Al}_2\text{Si}_6\text{O}_{18}$. Find n.

Numerical response: _____

Q74.

In the molecular dimer Al_2Cl_6 , how many chlorine atoms act as bridges between the two aluminium atoms?

Numerical response: _____

SECTION 7 — INTEGRATED QUESTIONS (Q75-Q85)

These questions mix reactions, structures, statements and numerical ideas. Read each question carefully and answer in the format given.

Q75.

Consider the following statements.

1. NH_3 , MeNH_2 and Me_2NH are grouped as unsymmetrical-cleavage donors.
2. Tertiary amines can give symmetrical neutral $\text{BH}_3 \cdot \text{L}$ adducts.
3. NaH can give a BH_4^- salt through hydride attack.
4. Bridge B–H contacts are longer and weaker than terminal B–H bonds.
5. CO is classified as an unsymmetrical amine-type cleavage reagent in the supplied scheme.
6. Primary, secondary and tertiary amines must give identical products with diborane.

Which option gives the complete set of correct statements?

- 1) 1, 2, 3, 5
- 2) 1, 2, 4, 6
- 3) 1, 2, 3, 4
- 4) 1, 3, 4, 5

Q76.

Consider the following statements.

1. Sodium peroxoborate is associated with active-oxygen chemistry.
2. In alkaline medium it can give H_2O_2 -related oxidising species.
3. It can oxidise Cr(III) to Cr(VI).
4. Cr(VI) is represented mainly as chromate in alkaline solution.
5. Peroxoborate acts only as a reducing agent in alkaline medium.
6. Its oxidation chemistry is unrelated to peroxide or active oxygen.

Which option gives the complete set of correct statements?

- 1) 1, 2, 3, 5
- 2) 1, 2, 3, 4
- 3) 1, 2, 4, 6
- 4) 1, 3, 4, 5

Q77.

Consider the following statements.

1. Hydrolysis of BF_3 generates HF.
2. HF attacks the silica network of glass.
3. Boric acid is the boron-containing hydrolysis product.
4. Moisture therefore changes the practical compatibility of BF_3 with glass.
5. Hydrolysis of BCl_3 produces HF as the principal hydrogen halide.
6. Silica is completely inert toward aqueous HF.

Which option gives the complete set of correct statements?

- 1) 1, 2, 3, 5
- 2) 1, 2, 4, 6
- 3) 1, 3, 4, 5
- 4) 1, 2, 3, 4

Q78.

Consider the following statements.

1. Cobalt compounds can give a characteristic blue borax bead.
2. Chromium compounds can give a characteristic green bead.
3. Manganese compounds can give a violet or amethyst bead under suitable conditions.
4. Copper compounds can give a characteristic coloured bead.
5. Ca^{2+} is expected to give the most diagnostic transition-metal-type coloured bead of the set.
6. The colours arise from metal borate or metaborate species in the glassy bead.

Which option gives the complete set of correct statements?

- 1) 1, 2, 3, 4, 6
- 2) 1, 2, 3, 4, 5
- 3) 1, 2, 3, 5, 6
- 4) 1, 2, 4, 5, 6

Q79.

Consider the following statements.

1. Borax can be acidified to obtain boric acid.
2. Thermal dehydration of boric acid can give B_2O_3 .
3. Reduction of B_2O_3 with Mg gives crude boron.
4. Acid washing helps remove magnesium-containing impurities.
5. Direct reduction of borax with water is the standard route to elemental boron.
6. High-temperature reduction of BBr_3 with H_2 can give purer boron.

Which option gives the complete set of correct statements?

- 1) 1, 2, 3, 4, 5
- 2) 1, 2, 3, 5, 6
- 3) 1, 2, 3, 4, 6
- 4) 1, 2, 4, 5, 6

Q80.

Select all correct statements.

- 1) Al_2O_3 is dissolved in a molten fluoride bath containing cryolite.
- 2) The bath lowers the effective operating temperature and improves conductivity.
- 3) Carbon or graphite electrodes are used, and carbon is consumed at the anode.
- 4) Aluminium is produced at the cathodic region.
- 5) Electrolysis of aqueous AlCl_3 is the standard industrial route to aluminium metal.
- 6) Aluminium metal is liberated at the carbon anode.

Q81.

How many are correct? (i) concentrated HNO_3 can be stored in Al because of passivation; (ii) Al wire is used in transmission; (iii) Al + NaOH can be used in drain-cleaning formulations; (iv) washing soda can attack Al by forming soluble aluminate; (v) $\text{Al}(\text{OH})_3$ is used as an antacid; (vi) keeping water overnight in bare Al utensils is presented as the recommended statement in the question bank.

Answer: _____

Q82.

How many statements are correct? (i) BeO and Al_2O_3 are amphoteric; (ii) Be_2C and Al_4C_3 give CH_4 on hydrolysis; (iii) Be/Al nitrides give NH_3 on hydrolysis; (iv) Be/Al chlorides show association/bridging; (v) B_2O_3 and SiO_2 are acidic oxides; (vi) BCl_3 and SiCl_4 are covalent and hydrolysable; (vii) B and Si form hard refractory carbide/boride-type compounds; (viii) B and Si must have identical formulae for every hydride and halide.

Answer: _____

Q83.

How many statements are correct?

1. TiI_3 is better interpreted through $\text{Ti}(\text{I})$ and a polyiodide ion than as simple $\text{Ti}^{3+} + 3\text{I}^-$.
2. Boric acid is a weak monobasic Lewis acid.
3. Hydrated borax contains two trigonal and two tetrahedral boron centres.

4. The conventional borax framework contains five B-O-B linkages and no B=O π bond in the localised model.
5. B_2H_6 contains both 2c-2e terminal bonds and 3c-2e bridge bonds.
6. NH_3 /primary/secondary amines and tertiary donors can give different cleavage patterns of B_2H_6 .
7. Borazine is called inorganic benzene but is not identical to benzene in chemical reactivity.
8. $\text{Al}_2(\text{CH}_3)_6$ is an electron-deficient bridged dimer.
9. Dry AlCl_3 and hydrated aluminium species need not have the same bonding description.
10. AlCl_3 can be associated/bridged, and its hydrolysis explains white fumes in moist air.
11. Lewis acidity of BX_3 in the familiar donor comparison increases from BF_3 toward BI_3 .
12. Boron can be obtained from borax through H_3BO_3 and B_2O_3 followed by reduction.
13. Aluminium is protected by passivation in concentrated HNO_3 but is attacked by strong alkali.
14. Every Group 13 halide is a monomeric trigonal molecule in every physical state.
15. All six hydrogens in B_2H_6 are equivalent ordinary 2c-2e terminal hydrogens.

Answer: _____

Q84.

(a) Identify the incorrect reaction:

- A. $\text{BF}_3 + \text{F}^- \rightarrow [\text{BF}_4]^-$
 - B. $\text{BF}_3 + 2\text{H}_2\text{O} \rightarrow [\text{BF}_3\text{OH}]^- + \text{H}_3\text{O}^+$
 - C. $\text{BCl}_3 + 3\text{EtOH} \rightarrow \text{B}(\text{OEt})_3 + 3\text{HCl}$
 - D. $\text{BCl}_3 + 2\text{C}_5\text{H}_5\text{N} \rightarrow \text{Cl}_3\text{B}(\text{C}_5\text{H}_5\text{N})_2$
- (b) How many statements are scientifically correct?
- A. All boron trihalides are hydrolysed completely.
 - B. Anhydrous AlCl_3 is ionic.
 - C. AlCl_3 is a weaker Lewis acid than BCl_3 .
 - D. The coordination number of Al in dimeric AlCl_3 is 6.

Answer(s): _____

Q85.

- (a) For $3\text{B}_2\text{H}_6 + 6\text{NH}_3$ at room temperature $\rightarrow$ 3 mol ionic product A; on heating, A $\rightarrow$ product B + flammable gas C. Identify A, B and C. State the atomic number of the element in gas C and the number of NH_3 molecules in the cationic part of one formula unit of A.
- (b) For H^+ , benzene, H_2O , CH_3^- , B_2H_6 , CH_3^+ , $(\text{C}_2\text{H}_5)_2\text{O}$, PH_3 , acetone, H^- , CO and NO, count and identify the Lewis acids and the Lewis bases.

Answer(s): _____

SECTION 8 — COUNT THE CORRECT STATEMENTS (Q86-Q90)

Five additional statement-count questions integrating J.D. Lee, PMS, VKJ and NCERT source material on peroxoborate, boron flame tests, borazine addition chemistry, mixed-valence heavier halides, and applied aluminium/boron chemistry. Enter the number of correct statements.

Q86.

How many statements are correct?

1. Sodium peroxoborate is commonly written as $\text{NaBO}_3 \cdot 4\text{H}_2\text{O}$.
2. Structurally it is the dimeric salt $\text{Na}_2[(\text{HO})_2\text{B}(\text{O})_2\text{B}(\text{OH})_2] \cdot 6\text{H}_2\text{O}$.
3. Its anion contains two genuine peroxide ($-\text{O}-\text{O}-$) bridges linking two boron centres.
4. Both boron atoms in the anion are four-coordinate (tetrahedral).
5. It can be prepared by electrolysis of a sodium borate solution or by treating sodium metaborate with H_2O_2 .
6. On warming above about 80°C the O-O linkages break, releasing H_2O_2 .
7. It is used as an oxygen-releasing bleach and brightener in detergents.
8. In alkaline solution it can oxidise Cr(III) to chromate, CrO_4^{2-} .
9. It is a convenient solid source of hydrogen peroxide / active oxygen.
10. The prefix “peroxo” is only nominal; the compound has no O-O bond.
11. Sodium metaborate, NaBO_2 , contains the same peroxide linkage.

12. Sodium peroxoborate acts only as a reducing agent and never as an oxidant.

Number of correct statements: _____

Q87.

How many statements are correct?

1. Warming a borate with conc. H_2SO_4 and CaF_2 releases BF_3 , which colours a flame green.
2. Warming a borate with methanol and conc. H_2SO_4 gives volatile trimethyl borate, $\text{B}(\text{OCH}_3)_3$, which burns with a green flame.
3. In the methyl-borate test the conc. H_2SO_4 removes water and drives ester formation.
4. Turmeric (curcumin) paper moistened with an acidified borate solution turns red to reddish-brown (rosocyanine).
5. This red turmeric colour turns greenish-black on exposure to ammonia.
6. The green flame colour is produced by a volatile covalent boron species, not by a free boron ion.
7. Both flame tests require the boron first to be converted to boric acid.
8. Moist BF_3 can attack the silica of glassware because its hydrolysis generates HF.
9. Boron itself gives an intense crimson atomic flame colour similar to lithium.
10. Perfectly dry BF_3 rapidly etches silica by an aqueous-HF mechanism.
11. Trimethyl borate is non-volatile, so it cannot be used for a flame test.
12. A large amount of sodium in the sample can mask the green flame with its own yellow colour.

Number of correct statements: _____

Q88.

How many statements are correct?

1. Borazine, $\text{B}_3\text{N}_3\text{H}_6$, is a planar six-membered ring with alternating B and N atoms.
2. It is isoelectronic with benzene and is called "inorganic benzene".
3. In borazine boron is the δ^+ centre and nitrogen the δ^- centre.
4. Unlike benzene, borazine readily undergoes addition reactions across the B-N bonds.
5. Borazine adds 3 HCl to give $\text{B}_3\text{N}_3\text{H}_9\text{Cl}_3$, chlorine adding to boron and hydrogen to nitrogen.
6. Borazine can be hydrogenated to $\text{B}_3\text{N}_3\text{H}_{12}$.
7. Hydrolysis of borazine yields boric acid, ammonia and hydrogen.
8. Borazine can be prepared by heating the diammoniate of diborane, or B_2H_6 with excess NH_3 .
9. B-trichloroborazine, $\text{B}_3\text{N}_3\text{H}_3\text{Cl}_3$, is obtained from 3 BCl_3 + 3 NH_4Cl with elimination of 9 HCl.
10. B-trichloroborazine can be reduced by NaBH_4 to borazine.
11. Borazine is aromatically more stable than benzene and therefore resists addition reactions.
12. Borazine and benzene are equally resistant to hydrolysis.

Number of correct statements: _____

Q89.

How many statements are correct?

1. " TlI_3 " is best written as $\text{Tl}^+(\text{I}_3^-)$, thallium remaining in the +1 state.
2. " GaCl_2 " is a mixed-valence compound formulated as $\text{Ga}^+[\text{GaCl}_4]^-$.
3. " InCl_2 " is similarly $\text{In}^+[\text{InCl}_4]^-$ rather than true $\text{In}(\text{II})$.
4. TlI_3 is isomorphous with CsI_3 and NH_4I_3 and contains a linear I_3^- ion.
5. $\text{Tl}(\text{I})$ resembles $\text{K}(\text{I})$: TlOH is a water-soluble strong base.
6. $\text{Tl}(\text{I})$ also resembles $\text{Ag}(\text{I})$: TlCl , TlBr and TlI are sparingly soluble and photosensitive.
7. Tl_2S is black, paralleling Ag_2S .
8. Thallium compounds are cumulative poisons.
9. The stability of the +1 state increases down Group 13 as $\text{Al} < \text{Ga} < \text{In} < \text{Tl}$.
10. In " GaCl_2 " gallium is genuinely +2, giving a paramagnetic species with one unpaired electron.
11. TlI_3 is correctly formulated as Tl^{3+} with three independent I^- ions.
12. B_2Cl_4 contains a real B-B bond and adopts a non-eclipsed conformation in the gas phase.

Number of correct statements: _____

Q90.

How many statements are correct?

1. Anodising thickens the Al_2O_3 film by making the article the anode in dilute H_2SO_4 ; the porous oxide can then be dyed.
2. Anhydrous AlCl_3 and BF_3 catalyse Friedel-Crafts alkylation and acylation.
3. In Friedel-Crafts reactions AlCl_3 is not strictly catalytic because $[\text{AlCl}_4]^-$ is formed and consumed.
4. Fluoroboric acid, HBF_4 , is a strong acid and contains the tetrahedral $[\text{BF}_4]^-$ ion.
5. KBF_4 , like KClO_4 , is only sparingly soluble in water.
6. ^{10}B has a high absorption cross-section for thermal neutrons, so boron and boron steel are used in control rods and shielding.
7. Boron carbide (B_4C , more truly B_{13}C_2) is a very hard abrasive used in bullet-proof clothing.
8. Gallium arsenide and gallium nitride are semiconductors used in LEDs and laser diodes.
9. $\text{Al}(\text{BH}_4)_3$ and $\text{Be}(\text{BH}_4)_2$ are covalent, volatile borohydrides in which BH_4 groups bridge the metal through three-centre bonds.
10. Anodising is carried out with the aluminium article as the cathode in molten cryolite.
11. The aluminothermic (thermite) reduction works because Al_2O_3 has a less negative enthalpy of formation than most metal oxides.
12. In Friedel-Crafts acylation the true active species is neutral AlCl_3 , recovered unchanged at the end.

Number of correct statements: _____

SECTION 9 — Complete the reaction — write the product(s)

155 selected reactions in the Boron-Family syllabus, grouped by topic after the requested deletions. Write the product(s); balance only where coefficients are shown; equilibria carry $\rightleftharpoons$. A complete reaction answer key follows.

A. Elemental boron — preparation

1. $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}(\text{aq}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow$ _____
2. $\text{Ca}_2\text{B}_6\text{O}_{11} + 4\text{SO}_2 + 11\text{H}_2\text{O} \rightarrow$ _____
3. $2\text{H}_3\text{BO}_3 \Delta \rightarrow$ _____
4. $\text{B}_2\text{O}_3 + 3\text{Mg} \Delta \rightarrow$ _____
5. $\text{B}_2\text{O}_3 + 6\text{Na} \Delta \rightarrow$ _____
6. $\text{B}_2\text{O}_3 + 6\text{K} \Delta \rightarrow$ _____
7. $\text{B}_2\text{O}_3 + 2\text{Al} \Delta \rightarrow$ _____
8. $2\text{BCl}_3(\text{g}) + 3\text{H}_2(\text{g}) \text{ hot W/Ta} \rightarrow$ _____
9. $2\text{BI}_3(\text{g}) \text{ hot filament} \rightarrow$ _____
10. $\text{B}_2\text{H}_6(\text{g}) \Delta (\text{pyrolysis}) \rightarrow$ _____

B. Elemental boron — reactions

11. $4\text{B} + 3\text{O}_2 \Delta \rightarrow$ _____
12. $2\text{B} + 3\text{Na}_2\text{O}_2 \text{ fusion} \rightarrow$ _____
13. $2\text{B} + \text{N}_2 \text{ very high T} \rightarrow$ _____
14. $2\text{B} + 2\text{NH}_3 \text{ very high T} \rightarrow$ _____
15. $2\text{B} + 3\text{Cl}_2 \Delta \rightarrow$ _____
16. $2\text{B} + 3\text{S } 1200^\circ\text{C} \rightarrow$ _____
17. $4\text{B} + \text{C} \Delta \rightarrow$ _____
18. $2\text{B} + 3\text{Mg} \Delta \rightarrow$ _____
19. $2\text{B} + 6\text{NaOH}(\text{l}) \text{ fused} \rightarrow$ _____
20. $4\text{B} + 4\text{NaOH} + 3\text{O}_2 \text{ fused, air} \rightarrow$ _____
21. $\text{B} + 3\text{HNO}_3(\text{conc.}) \rightarrow$ _____
22. $2\text{B} + 3\text{H}_2\text{SO}_4(\text{conc.}) \rightarrow$ _____

C. Aluminium — reactions

23. $4\text{Al} + 3\text{O}_2 \rightarrow$ _____
24. $2\text{Al} + \text{N}_2 \Delta \rightarrow$ _____
25. $\text{AlN} + 3\text{H}_2\text{O} \rightarrow$ _____
26. $2\text{Al} + 6\text{H}_2\text{O} (\text{hot, oxide film removed}) \rightarrow$ _____
27. $2\text{Al} + 3\text{H}_2\text{O}(\text{g}) \text{ steam} \rightarrow$ _____
28. $2\text{Al} + 6\text{HCl}(\text{aq}) \rightarrow$ _____
29. $2\text{Al} + 3\text{H}_2\text{SO}_4(\text{dil.}) \rightarrow$ _____
30. $2\text{Al} + 6\text{H}_2\text{SO}_4(\text{conc.}) \Delta \rightarrow$ _____
31. $2\text{Al} + 2\text{NaOH} + 6\text{H}_2\text{O} \rightarrow$ _____
32. $2\text{Al} + 2\text{NaOH} + 2\text{H}_2\text{O} \text{ fused} \rightarrow$ _____
33. $2\text{Al} + 3\text{Cl}_2 \text{ dry, } \Delta \rightarrow$ _____
34. $4\text{Al} + 3\text{C} \Delta \rightarrow$ _____
35. $\text{Al}_4\text{C}_3 + 12\text{H}_2\text{O} \rightarrow$ _____

D. Gallium, indium, thallium

36. $2\text{Ga} + 2\text{NaOH} + 6\text{H}_2\text{O} \rightarrow \underline{\hspace{2cm}}$
37. 3GaX(aq) disproportionation $\rightarrow \underline{\hspace{2cm}}$
38. 3InX(aq) disproportionation $\rightarrow \underline{\hspace{2cm}}$
39. $\text{GaCl}_3(\text{l}) + \text{Ga(s)} \rightarrow \underline{\hspace{2cm}}$
40. $\text{In} + 2\text{HCl(g)} \rightarrow \underline{\hspace{2cm}}$
41. $\text{TlI} + \text{I}_2 \rightarrow \underline{\hspace{2cm}}$
42. $\text{TlF} + \text{F}_2 \rightarrow \underline{\hspace{2cm}}$
43. $2[\text{Tl}^{3+}] + \text{reducing I}^- : \text{Tl}^{3+} + 2\text{e}^- \rightleftharpoons \underline{\hspace{2cm}}$

E. Oxides and hydroxides

44. $\text{B}_2\text{O}_3 + 3\text{H}_2\text{O} \rightarrow \underline{\hspace{2cm}}$
45. $\text{B}_2\text{O}_3 + 3\text{CaO} \rightarrow \underline{\hspace{2cm}}$
46. $\text{B}_2\text{O}_3 + \text{CaO} \rightarrow \underline{\hspace{2cm}}$
47. $\text{B}_2\text{O}_3 + 2\text{NaOH(l)} \rightarrow \underline{\hspace{2cm}}$
48. $\text{B}_2\text{O}_3 + \text{P}_2\text{O}_5 \rightarrow \underline{\hspace{2cm}}$
49. $\text{B}_2\text{O}_3(\text{l}) + \text{CuO(s)}$ bead $\rightarrow \underline{\hspace{2cm}}$
50. $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow \underline{\hspace{2cm}}$
51. $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow \underline{\hspace{2cm}}$
52. $\text{Al(OH)}_3 + 3\text{H}^+ \rightarrow \underline{\hspace{2cm}}$
53. $\text{Al(OH)}_3 + \text{OH}^- \rightarrow \underline{\hspace{2cm}}$
54. $\text{Al}^{3+}(\text{aq}) + 3\text{NH}_3 + 3\text{H}_2\text{O} \rightarrow \underline{\hspace{2cm}}$
55. $[\text{Al(OH)}_4]^- + \text{CO}_2 \rightarrow \underline{\hspace{2cm}}$
56. $2\text{Al(OH)}_3 \Delta \rightarrow \underline{\hspace{2cm}}$
57. $\text{Al}_2(\text{SO}_4)_3$ strong $\Delta \rightarrow \underline{\hspace{2cm}}$
58. $(\text{NH}_4)_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O} \Delta \rightarrow \underline{\hspace{2cm}}$

F. Boric acid, H_3BO_3

59. $\text{B(OH)}_3 + 2\text{H}_2\text{O} \rightleftharpoons \underline{\hspace{2cm}}$
60. $\text{BX}_3 + 3\text{H}_2\text{O} \rightarrow \underline{\hspace{2cm}}$
61. $\text{BH}_3 + 3\text{H}_2\text{O} \rightarrow \underline{\hspace{2cm}}$
62. $\text{H}_3\text{BO}_3 \approx 370 \text{ K} \rightarrow \underline{\hspace{2cm}}$
63. $4\text{HBO}_2 \Delta \rightarrow \underline{\hspace{2cm}}$
64. $\text{H}_2\text{B}_4\text{O}_7$ red heat $\rightarrow \underline{\hspace{2cm}}$
65. $\text{H}_3\text{BO}_3 + 4\text{HF} \rightarrow \underline{\hspace{2cm}}$
66. $\text{B(OH)}_3 + 3\text{CH}_3\text{OH} \rightleftharpoons \underline{\hspace{2cm}}$
67. $\text{H}_3\text{BO}_3 + 3\text{C}_2\text{H}_5\text{OH conc. H}_2\text{SO}_4 \rightarrow \underline{\hspace{2cm}}$
68. $2\text{H}_3\text{BO}_3 + \text{Na}_2\text{CO}_3(\text{hot}) \rightarrow \underline{\hspace{2cm}}$

G. Borax and borates

69. $\text{Ca}_2\text{B}_6\text{O}_{11} \cdot 5\text{H}_2\text{O} + 2\text{Na}_2\text{CO}_3 + 5\text{H}_2\text{O} \rightarrow \underline{\hspace{2cm}}$
70. $4\text{NaBO}_2 + \text{CO}_2 \rightarrow \underline{\hspace{2cm}}$
71. $\text{Na}_2\text{B}_4\text{O}_7 + 7\text{H}_2\text{O}$ hydrolysis $\rightarrow \underline{\hspace{2cm}}$
72. $[\text{B}_4\text{O}_5(\text{OH})_4]^{2-} + 5\text{H}_2\text{O} \rightleftharpoons \underline{\hspace{2cm}}$
73. $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O} + 2\text{HCl} \rightarrow \underline{\hspace{2cm}}$
74. $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$ heat, $-\text{H}_2\text{O} \rightarrow \underline{\hspace{2cm}}$
75. $\text{Na}_2\text{B}_4\text{O}_7$ strong heat $\rightarrow \underline{\hspace{2cm}}$
76. $\text{Na}_2\text{B}_4\text{O}_7 + 6\text{CaF}_2 + 7\text{H}_2\text{SO}_4 \rightarrow \underline{\hspace{2cm}}$
77. $\text{CoO} + \text{B}_2\text{O}_3$ bead $\rightarrow \underline{\hspace{2cm}}$

H. Sodium peroxoborate (perborate)

78. $2\text{NaBO}_2 + 2\text{H}_2\text{O}_2 + 6\text{H}_2\text{O} \rightarrow \underline{\hspace{2cm}}$
79. $\text{NaBO}_3 \cdot 4\text{H}_2\text{O(aq)} \xrightarrow{>80^\circ\text{C}} \underline{\hspace{2cm}}$

I. Boron trihalides and fluoroboric acid

80. $\text{B}_2\text{O}_3 + 3\text{CaF}_2 + 3\text{H}_2\text{SO}_4(\text{conc.}) \rightarrow \underline{\hspace{2cm}}$
81. $\text{B}_2\text{O}_3 + 6\text{NH}_4\text{BF}_4 \Delta \rightarrow \underline{\hspace{2cm}}$
82. $\text{NH}_4\text{BF}_4 \Delta \rightarrow \underline{\hspace{2cm}}$
83. $\text{B}_2\text{O}_3 + 3\text{C} + 3\text{Cl}_2 \Delta \rightarrow \underline{\hspace{2cm}}$
84. $\text{BF}_3 + \text{NH}_3 \rightarrow \underline{\hspace{2cm}}$
85. $\text{BF}_3 + (\text{C}_2\text{H}_5)_2\text{O} \rightarrow \underline{\hspace{2cm}}$
86. $\text{BF}_3 + \text{F}^- \rightarrow \underline{\hspace{2cm}}$
87. $\text{BCl}_3 + \text{:NH}_3 \rightarrow \underline{\hspace{2cm}}$
88. $\text{BCl}_3 + \text{AsH}_3$ selective $\rightarrow \underline{\hspace{2cm}}$
89. $4\text{BF}_3 + 3\text{H}_2\text{O} \rightarrow \underline{\hspace{2cm}}$
90. $\text{BCl}_3 + 3\text{H}_2\text{O} \rightarrow \underline{\hspace{2cm}}$
91. $\text{BF}_3 + \text{BCl}_3$ redistribution $\rightarrow \underline{\hspace{2cm}}$
92. $\text{H}_3\text{BO}_3 + 3\text{HF} \rightarrow \underline{\hspace{2cm}}$

J. Diborane — preparation

93. $\text{Mg}_3\text{B}_2 + 6\text{H}_3\text{PO}_4 \rightarrow \underline{\hspace{2cm}}$
94. $2\text{BF}_3 + 6\text{NaH}$ 450 K $\rightarrow \underline{\hspace{2cm}}$
95. $4\text{BF}_3 \cdot \text{OEt}_2 + 3\text{LiAlH}_4$ dry ether $\rightarrow \underline{\hspace{2cm}}$
96. $4\text{BCl}_3 + 3\text{LiAlH}_4$ ether $\rightarrow \underline{\hspace{2cm}}$
97. $2\text{NaBH}_4 + \text{I}_2$ diglyme $\rightarrow \underline{\hspace{2cm}}$

98. $3\text{NaBH}_4 + 4\text{BF}_3 \cdot \text{OEt}_2 \rightarrow \underline{\hspace{2cm}}$
99. $2\text{BCl}_3 + 6\text{H}_2$ electric discharge $\rightarrow \underline{\hspace{2cm}}$
100. $\text{B}_2\text{O}_3 + 2\text{Al} + 3\text{H}_2 \Delta \rightarrow \underline{\hspace{2cm}}$

K. Diborane — reactions and cleavage

101. $\text{B}_2\text{H}_6 + 3\text{O}_2 \rightarrow \underline{\hspace{2cm}}$
102. $\text{B}_2\text{H}_6 + 6\text{H}_2\text{O} \rightarrow \underline{\hspace{2cm}}$
103. $\text{B}_2\text{H}_6 + 6\text{NaOH} \rightarrow \underline{\hspace{2cm}}$
104. $\text{B}_2\text{H}_6 + 6\text{CH}_3\text{OH} \rightarrow \underline{\hspace{2cm}}$
105. $\text{B}_2\text{H}_6 + 3\text{Cl}_2 \rightarrow \underline{\hspace{2cm}}$
106. $\text{B}_2\text{H}_6 + \text{HCl} \rightarrow \underline{\hspace{2cm}}$
107. $\text{B}_2\text{H}_6 + 2\text{LiH} \rightarrow \underline{\hspace{2cm}}$
108. $2\text{B}_2\text{H}_6 + 2\text{Na} \rightarrow \underline{\hspace{2cm}}$
109. $\text{B}_2\text{H}_6 + 2\text{CO} \Delta, \text{p} \rightarrow \underline{\hspace{2cm}}$
110. $\text{B}_2\text{H}_6 + 2\text{PF}_3 \rightarrow \underline{\hspace{2cm}}$
111. $\text{B}_2\text{H}_6 + 2\text{P(CH}_3)_3 \rightarrow \underline{\hspace{2cm}}$
112. $\text{B}_2\text{H}_6 + 2\text{N(CH}_3)_3 \rightarrow \underline{\hspace{2cm}}$
113. $\text{B}_2\text{H}_6 + 2\text{Et}_2\text{O} \rightarrow \underline{\hspace{2cm}}$
114. $\text{B}_2\text{H}_6 + 2\text{Et}_2\text{S} \rightarrow \underline{\hspace{2cm}}$
115. $\text{B}_2\text{H}_6 + 2\text{NH}_3$ low T $\rightarrow \underline{\hspace{2cm}}$
116. $3\text{B}_2\text{H}_6 + 6\text{NH}_3 \Delta \rightarrow \underline{\hspace{2cm}}$

L. Borohydrides and complex hydrides

117. $4\text{NaH} + \text{B(OCH}_3)_3 \approx 250^\circ\text{C} \rightarrow \underline{\hspace{2cm}}$
118. $2\text{MH} + \text{B}_2\text{H}_6$ ether (M = Li, Na) $\rightarrow \underline{\hspace{2cm}}$
119. $\text{NaBH}_4 + \text{LiCl} \rightarrow \underline{\hspace{2cm}}$
120. $\text{BeCl}_2 + 2\text{LiBH}_4 \rightarrow \underline{\hspace{2cm}}$
121. $3\text{LiBH}_4 + \text{AlCl}_3 \rightarrow \underline{\hspace{2cm}}$
122. $4\text{LiH} + \text{AlCl}_3$ dry ether $\rightarrow \underline{\hspace{2cm}}$
123. $\text{LiBH}_4 + 2\text{H}_2\text{O} \rightarrow \underline{\hspace{2cm}}$

M. Borazine and boron nitride

124. $3\text{NH}_4\text{Cl} + 3\text{NaBH}_4 \Delta \rightarrow \underline{\hspace{2cm}}$
125. $3\text{BCl}_3 + 3\text{NH}_4\text{Cl}$ 140 $^\circ\text{C} \rightarrow \underline{\hspace{2cm}}$
126. $\text{B}_3\text{N}_3\text{H}_3\text{Cl}_3 + 3\text{NaBH}_4 \rightarrow \underline{\hspace{2cm}}$
127. $\text{B}_3\text{N}_3\text{H}_6 + 3\text{HCl} \rightarrow \underline{\hspace{2cm}}$
128. $\text{B}_3\text{N}_3\text{H}_6 + 9\text{H}_2\text{O} \rightarrow \underline{\hspace{2cm}}$
129. $\text{B}_3\text{N}_3\text{H}_6$ strong $\Delta \rightarrow \underline{\hspace{2cm}}$
130. $\text{B}_2\text{O}_3 + 2\text{NH}_3 \Delta \rightarrow \underline{\hspace{2cm}}$
131. $\text{BCl}_3 + \text{NH}_3 \Delta \rightarrow \underline{\hspace{2cm}}$

N. Boron carbide, B_4C

132. $2\text{B}_2\text{O}_3 + 7\text{C} \approx 2500^\circ\text{C} \rightarrow \underline{\hspace{2cm}}$

O. Aluminothermy (thermite / Goldschmidt)

133. $\text{Fe}_2\text{O}_3 + 2\text{Al} \Delta \rightarrow \underline{\hspace{2cm}}$
134. $\text{Cr}_2\text{O}_3 + 2\text{Al} \Delta \rightarrow \underline{\hspace{2cm}}$
135. $3\text{Mn}_3\text{O}_4 + 8\text{Al} \Delta \rightarrow \underline{\hspace{2cm}}$

P. Aluminium chloride and aqueous aluminium chemistry

136. $\text{Al}_2\text{O}_3 + 3\text{C} + 3\text{Cl}_2 \Delta \rightarrow \underline{\hspace{2cm}}$
137. $2\text{AlCl}_3(\text{g}) \rightleftharpoons \underline{\hspace{2cm}}$
138. $\text{Al}_2\text{Cl}_6 + 12\text{H}_2\text{O} \rightarrow \underline{\hspace{2cm}}$
139. $2\text{Al} + 6\text{HCl(aq)} + 12\text{H}_2\text{O} \rightarrow \underline{\hspace{2cm}}$
140. $\text{Al(OH)}_3 + 3\text{HCl} + 3\text{H}_2\text{O} \rightarrow \underline{\hspace{2cm}}$
141. $2\text{AlCl}_3 \cdot 6\text{H}_2\text{O} \Delta \rightarrow \underline{\hspace{2cm}}$
142. $\text{AlCl}_3 + 6\text{NH}_3 \rightarrow \underline{\hspace{2cm}}$
143. $\text{AlCl}_3(\text{aq}) + 3\text{NH}_4\text{OH} \rightarrow \underline{\hspace{2cm}}$
144. $\text{Al(OH)}_3 + \text{NaOH} \rightarrow \underline{\hspace{2cm}}$
145. $[\text{Al(H}_2\text{O)}_6]^{3+} + \text{H}_2\text{O} \rightleftharpoons \underline{\hspace{2cm}}$

Q. Aluminium sulfate and alums

146. $2\text{Al(OH)}_3 + 3\text{H}_2\text{SO}_4 \rightarrow \underline{\hspace{2cm}}$
147. $\text{Al}_2(\text{SO}_4)_3 + \text{K}_2\text{SO}_4 + 24\text{H}_2\text{O} \rightarrow \underline{\hspace{2cm}}$
148. $\text{KAl(SO}_4)_2 \cdot 12\text{H}_2\text{O(s)}$ dissolve $\rightarrow \underline{\hspace{2cm}}$
149. $\text{KAl(SO}_4)_2 \cdot 12\text{H}_2\text{O} \Delta$ (burnt alum) $\rightarrow \underline{\hspace{2cm}}$

R. Aluminium extraction — Hall-Héroult electrode reactions

150. Cathode: $\text{Al}^{3+} + 3\text{e}^- \rightarrow \underline{\hspace{2cm}}$
151. Anode: $\text{C} + 2\text{O}_2^- \rightarrow \underline{\hspace{2cm}}$
152. Overall: $2\text{Al}_2\text{O}_3 + 3\text{C} \rightarrow \underline{\hspace{2cm}}$
153. Bayer: $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow \underline{\hspace{2cm}}$
154. Bayer: $\text{Na[Al(OH)}_4]$ seeding/dilution $\rightarrow \underline{\hspace{2cm}}$
155. 2Al(OH)_3 calcination $\rightarrow \underline{\hspace{2cm}}$

BORON FAMILY - ANSWER KEY (Q1-Q90)

Q.	Answer	Q.	Answer	Q.	Answer	Q.	Answer
NCERT CORE - Q1-Q16							
Q1	13	Q5	12	Q9	15	Q13	14 - all statements correct
Q2	14	Q6	12	Q10	11	Q14	15
Q3	18	Q7	11	Q11	11	Q15	13
Q4	14	Q8	12	Q12	11	Q16	13
SINGLE-CORRECT OBJECTIVE - Q17-Q31							
Q17	B	Q21	A	Q25	B	Q29	B
Q18	D	Q22	D	Q26	D	Q30	D
Q19	A	Q23	A	Q27	A	Q31	C
Q20	C	Q24	C	Q28	C		
ONE-OR-MORE-CORRECT - Q32-Q37							
Q32	A, B, C, D	Q34	A, B, C, D, E	Q36	A, B, C, D, E		
Q33	A, B, C, D, E	Q35	A, B, C, D, E	Q37	A, B, C, D		
MATRIX AND FLOW - Q38-Q47							
Q38	A-S, B-Q, C-R, D-P	Q41	A-P/S, B-Q/S, C-Q/S, D-R	Q44	Set I: A-S, B-T, C-R, D-P; Set II: A-S, B-R, C-P, D-Q	Q47	Set I: A-Q,S; B-P; C-S; D-Q. Set II: A-R; B-P; C-S; D-T; E-Q
Q39	A-Q, B-P, C-R/S, D-Q	Q42	A-P, B-Q, C-R, D-S	Q45	Set I: A-P,Q; B-R,T; C-P,S; D-P,Q. Set II: A-Q; B-R; C-S; D-P		
Q40	A-P, B-Q, C-R, D-S	Q43	A-P,R,S; B-P,Q,S; C-P,Q,S; D-P,R,S	Q46	A-Q,R,T; B-P,S,T; C-Q,R,S,T; D-P,R,S,T		
STATEMENT COUNT - Q48-Q54							
Q48	10	Q50	10	Q52	10	Q54	3; correct statements: 1, 2, 4
Q49	10	Q51	10	Q53	11		
NUMERICAL RESPONSE TYPE - Q55-Q74							
Q55	6	Q60	2	Q65	12	Q70	3 mol
Q56	4	Q61	5	Q66	1.5 mol	Q71	6 mol
Q57	2	Q62	14	Q67	6 mol	Q72	6
Q58	12	Q63	8	Q68	2 mol	Q73	3
Q59	2	Q64	6	Q69	2 mol	Q74	2
INTEGRATED TYPE QUESTIONS - Q75-Q85							
Q75	C	Q78	A	Q81	5	Q84	(a) D; (b) 0
Q76	B	Q79	C	Q82	7	Q85	(a) A=[BH ₂ (NH ₃) ₂]+[BH ₄]; B=B ₃ N ₃ H ₆ , C=H ₂ ; atomic number=1; NH ₃ in cation=2. (b) Lewis acids=3; Lewis bases=9
Q77	D	Q80	A, B, C, D	Q83	13		
EXTENDED STATEMENT COUNT - Q86-Q90							
Q86	9	Q88	10	Q90	9		
Q87	9	Q89	10				

SECTION 9 - REACTION ANSWER KEY

1. $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}(\text{aq}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow 4\text{H}_3\text{BO}_3(\text{s}) + \text{Na}_2\text{SO}_4(\text{aq}) + 5\text{H}_2\text{O}(\text{l})$
2. $\text{Ca}_2\text{B}_6\text{O}_{11} + 4\text{SO}_2 + 11\text{H}_2\text{O} \rightarrow 2\text{Ca}(\text{HSO}_3)_2 + 6\text{H}_3\text{BO}_3$
3. $2\text{H}_3\text{BO}_3 \Delta \rightarrow \text{B}_2\text{O}_3 + 3\text{H}_2\text{O}$
4. $\text{B}_2\text{O}_3 + 3\text{Mg} \Delta \rightarrow 2\text{B} + 3\text{MgO}$
5. $\text{B}_2\text{O}_3 + 6\text{Na} \Delta \rightarrow 2\text{B} + 3\text{Na}_2\text{O}$
6. $\text{B}_2\text{O}_3 + 6\text{K} \Delta \rightarrow 2\text{B} + 3\text{K}_2\text{O}$
7. $\text{B}_2\text{O}_3 + 2\text{Al} \Delta \rightarrow 2\text{B} + \text{Al}_2\text{O}_3$
8. $2\text{BCl}_3(\text{g}) + 3\text{H}_2(\text{g}) \text{ hot W/Ta} \rightarrow 2\text{B}(\text{s}) + 6\text{HCl}(\text{g})$
9. $2\text{BCl}_3(\text{g}) \text{ hot filament} \rightarrow 2\text{B}(\text{s}) + 3\text{Cl}_2(\text{g})$
10. $\text{B}_2\text{H}_6(\text{g}) \Delta \text{ (pyrolysis)} \rightarrow 2\text{B}(\text{s}) + 3\text{H}_2(\text{g})$
11. $4\text{B} + 3\text{O}_2 \Delta \rightarrow 2\text{B}_2\text{O}_3$
12. $2\text{B} + 3\text{Na}_2\text{O}_2 \text{ fusion} \rightarrow 2\text{Na}_3\text{BO}_3$
13. $2\text{B} + \text{N}_2 \text{ very high T} \rightarrow 2\text{BN}$
14. $2\text{B} + 2\text{NH}_3 \text{ very high T} \rightarrow 2\text{BN} + 3\text{H}_2$
15. $2\text{B} + 3\text{Cl}_2 \Delta \rightarrow 2\text{BCl}_3$
16. $2\text{B} + 3\text{S} \text{ 1200 }^\circ\text{C} \rightarrow \text{B}_2\text{S}_3$
17. $4\text{B} + \text{C} \Delta \rightarrow \text{B}_4\text{C}$
18. $2\text{B} + 3\text{Mg} \Delta \rightarrow \text{Mg}_3\text{B}_2$
19. $2\text{B} + 6\text{NaOH}(\text{l}) \text{ fused} \rightarrow 2\text{Na}_3\text{BO}_3 + 3\text{H}_2$
20. $4\text{B} + 4\text{NaOH} + 3\text{O}_2 \text{ fused, air} \rightarrow 4\text{NaBO}_2 + 2\text{H}_2\text{O}$
21. $\text{B} + 3\text{HNO}_3(\text{conc.}) \rightarrow \text{H}_3\text{BO}_3 + 3\text{NO}_2$
22. $2\text{B} + 3\text{H}_2\text{SO}_4(\text{conc.}) \rightarrow 2\text{H}_3\text{BO}_3 + 3\text{SO}_2$
23. $4\text{Al} + 3\text{O}_2 \rightarrow 2\text{Al}_2\text{O}_3$
24. $2\text{Al} + \text{N}_2 \Delta \rightarrow 2\text{AlN}$
25. $\text{AlN} + 3\text{H}_2\text{O} \rightarrow \text{Al}(\text{OH})_3 + \text{NH}_3$
26. $2\text{Al} + 6\text{H}_2\text{O} \text{ (hot, oxide film removed)} \rightarrow 2\text{Al}(\text{OH})_3 + 3\text{H}_2$
27. $2\text{Al} + 3\text{H}_2\text{O}(\text{g}) \text{ steam} \rightarrow \text{Al}_2\text{O}_3 + 3\text{H}_2$
28. $2\text{Al} + 6\text{HCl}(\text{aq}) \rightarrow 2\text{AlCl}_3(\text{aq}) + 3\text{H}_2$
29. $2\text{Al} + 3\text{H}_2\text{SO}_4(\text{dil.}) \rightarrow \text{Al}_2(\text{SO}_4)_3 + 3\text{H}_2$
30. $2\text{Al} + 6\text{H}_2\text{SO}_4(\text{conc.}) \Delta \rightarrow \text{Al}_2(\text{SO}_4)_3 + 3\text{SO}_2 + 6\text{H}_2\text{O}$
31. $2\text{Al} + 2\text{NaOH} + 6\text{H}_2\text{O} \rightarrow 2\text{Na}[\text{Al}(\text{OH})_4] + 3\text{H}_2$
32. $2\text{Al} + 2\text{NaOH} + 2\text{H}_2\text{O} \text{ fused} \rightarrow 2\text{NaAlO}_2 + 3\text{H}_2$
33. $2\text{Al} + 3\text{Cl}_2 \text{ dry, } \Delta \rightarrow 2\text{AlCl}_3 \rightarrow \text{Al}_2\text{Cl}_6$
34. $4\text{Al} + 3\text{C} \Delta \rightarrow \text{Al}_4\text{C}_3$
35. $\text{Al}_4\text{C}_3 + 12\text{H}_2\text{O} \rightarrow 4\text{Al}(\text{OH})_3 + 3\text{CH}_4$
36. $2\text{Ga} + 2\text{NaOH} + 6\text{H}_2\text{O} \rightarrow 2\text{Na}[\text{Ga}(\text{OH})_4] + 3\text{H}_2$
37. $3\text{GaX}(\text{aq}) \text{ disproportionation} \rightarrow 2\text{Ga} + \text{GaX}_3(\text{aq})$
38. $3\text{InX}(\text{aq}) \text{ disproportionation} \rightarrow 2\text{In} + \text{InX}_3(\text{aq})$
39. $\text{GaCl}_3(\text{l}) + \text{Ga}(\text{s}) \rightarrow 2\text{GaCl}_2 \text{ (i.e. Ga}^+[\text{GaCl}_4]^-)$
40. $\text{In} + 2\text{HCl}(\text{g}) \rightarrow \text{InCl}_2 + \text{H}_2 \text{ (i.e. In}^+[\text{InCl}_4]^-)$
41. $\text{Tl} + \text{I}_2 \rightarrow \text{Ti}^+[\text{I}_3]^-$
42. $\text{TlF} + \text{F}_2 \rightarrow \text{TlF}_3$
43. $2[\text{Ti}^{3+}] + \text{reducing I}^- : \text{Ti}^{3+} + 2\text{e}^- \rightleftharpoons \text{Ti}^+ \text{ (E}^\circ \approx +1.26 \text{ V)}$
44. $\text{B}_2\text{O}_3 + 3\text{H}_2\text{O} \rightarrow 2\text{H}_3\text{BO}_3$
45. $\text{B}_2\text{O}_3 + 3\text{CaO} \rightarrow \text{Ca}_3(\text{BO}_3)_2$
46. $\text{B}_2\text{O}_3 + \text{CaO} \rightarrow \text{Ca}(\text{BO}_2)_2$
47. $\text{B}_2\text{O}_3 + 2\text{NaOH}(\text{l}) \rightarrow 2\text{NaBO}_2 + \text{H}_2\text{O}$
48. $\text{B}_2\text{O}_3 + \text{P}_2\text{O}_5 \rightarrow 2\text{BP}_2\text{O}_4$
49. $\text{B}_2\text{O}_3(\text{l}) + \text{CuO}(\text{s}) \text{ bead} \rightarrow \text{Cu}(\text{BO}_2)_2 \text{ (blue glass)}$
50. $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$
51. $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{Na}[\text{Al}(\text{OH})_4]$
52. $\text{Al}(\text{OH})_3 + 3\text{H}^+ \rightarrow \text{Al}^{3+} + 3\text{H}_2\text{O}$
53. $\text{Al}(\text{OH})_3 + \text{OH}^- \rightarrow [\text{Al}(\text{OH})_4]^-$
54. $\text{Al}^{3+}(\text{aq}) + 3\text{NH}_3 + 3\text{H}_2\text{O} \rightarrow \text{Al}(\text{OH})_3 + 3\text{NH}_4^+$
55. $[\text{Al}(\text{OH})_4]^- + \text{CO}_2 \rightarrow \text{Al}(\text{OH})_3 + \text{HCO}_3^-$
56. $2\text{Al}(\text{OH})_3 \Delta \rightarrow \text{Al}_2\text{O}_3 + 3\text{H}_2\text{O}$
57. $\text{Al}_2(\text{SO}_4)_3 \text{ strong } \Delta \rightarrow \text{Al}_2\text{O}_3 + 3\text{SO}_3$
58. $(\text{NH}_4)_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O} \Delta \rightarrow \text{Al}_2\text{O}_3 + 2\text{NH}_3 + 4\text{SO}_3 + 25\text{H}_2\text{O}$
59. $\text{B}(\text{OH})_3 + \text{H}_2\text{O} \rightleftharpoons [\text{B}(\text{OH})_4]^- + \text{H}_3\text{O}^+$
60. $\text{B}_2\text{O}_3 + 3\text{H}_2\text{O} \rightarrow \text{H}_3\text{BO}_3 + 3\text{H}_2$
61. $\text{B}_2\text{O}_3 + 3\text{H}_2\text{O} \rightarrow \text{H}_3\text{BO}_3 + 3\text{H}_2$
62. $\text{H}_3\text{BO}_3 \approx 370 \text{ K} \rightarrow \text{HBO}_2 + \text{H}_2\text{O}$
63. $4\text{HBO}_2 \Delta \rightarrow \text{H}_2\text{B}_4\text{O}_7 + \text{H}_2\text{O}$
64. $\text{H}_2\text{B}_4\text{O}_7 \text{ red heat} \rightarrow 2\text{B}_2\text{O}_3 + \text{H}_2\text{O}$
65. $\text{H}_3\text{BO}_3 + 4\text{HF} \rightarrow \text{HBF}_4 + 3\text{H}_2\text{O}$
66. $\text{B}(\text{OH})_3 + 3\text{CH}_3\text{OH} \rightleftharpoons \text{B}(\text{OCH}_3)_3 + 3\text{H}_2\text{O}$
67. $\text{H}_3\text{BO}_3 + 3\text{C}_2\text{H}_5\text{OH conc. H}_2\text{SO}_4 \rightarrow \text{B}(\text{OC}_2\text{H}_5)_3 + 3\text{H}_2\text{O}$
68. $2\text{H}_3\text{BO}_3 + \text{Na}_2\text{CO}_3(\text{hot}) \rightarrow \text{Na}_2\text{B}_4\text{O}_7 + 6\text{H}_2\text{O} + \text{CO}_2 \text{ (write as 4H}_3\text{BO}_3 + \text{Na}_2\text{CO}_3)$
69. $\text{Ca}_2\text{B}_6\text{O}_{11} \cdot 5\text{H}_2\text{O} + 2\text{Na}_2\text{CO}_3 + 5\text{H}_2\text{O} \rightarrow \text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O} + 2\text{NaBO}_2 + 2\text{CaCO}_3$
70. $4\text{NaBO}_2 + \text{CO}_2 \rightarrow \text{Na}_2\text{B}_4\text{O}_7 + \text{Na}_2\text{CO}_3$
71. $\text{Na}_2\text{B}_4\text{O}_7 + 7\text{H}_2\text{O} \text{ hydrolysis} \rightarrow 2\text{NaOH} + 4\text{H}_3\text{BO}_3$
72. $[\text{B}_4\text{O}_5(\text{OH})_4]^{2-} + 5\text{H}_2\text{O} \rightleftharpoons 2\text{B}(\text{OH})_3 + 2[\text{B}(\text{OH})_4]^-$
73. $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O} + 2\text{HCl} \rightarrow 4\text{H}_3\text{BO}_3 + 2\text{NaCl} + 5\text{H}_2\text{O}$
74. $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O} \text{ heat, -H}_2\text{O} \rightarrow \text{Na}_2\text{B}_4\text{O}_7 \cdot 5\text{H}_2\text{O} + 5\text{H}_2\text{O}$
75. $\text{Na}_2\text{B}_4\text{O}_7 \text{ strong heat} \rightarrow 2\text{NaBO}_2 + \text{B}_2\text{O}_3$
76. $\text{Na}_2\text{B}_4\text{O}_7 + 6\text{CaF}_2 + 7\text{H}_2\text{SO}_4 \rightarrow 4\text{BF}_3 + 6\text{CaSO}_4 + \text{Na}_2\text{SO}_4 + 7\text{H}_2\text{O}$
77. $\text{CoO} + \text{B}_2\text{O}_3 \text{ bead} \rightarrow \text{Co}(\text{BO}_2)_2 \text{ (blue)}$
78. $2\text{NaBO}_2 + 2\text{H}_2\text{O}_2 + 6\text{H}_2\text{O} \rightarrow \text{Na}_2[\text{O}(\text{H})_2\text{B}(\text{O}-\text{O})_2\text{B}(\text{OH})_2] \cdot 6\text{H}_2\text{O}$
79. $\text{NaBO}_3 \cdot 4\text{H}_2\text{O}(\text{aq}) \rightarrow 80^\circ\text{C} \rightarrow \text{NaBO}_2 + \text{H}_2\text{O}_2 + 3\text{H}_2\text{O}$
80. $\text{B}_2\text{O}_3 + 3\text{CaF}_2 + 3\text{H}_2\text{SO}_4(\text{conc.}) \rightarrow 2\text{BF}_3 + 3\text{CaSO}_4 + 3\text{H}_2\text{O}$
81. $\text{B}_2\text{O}_3 + 6\text{NH}_4\text{BF}_4 \Delta \rightarrow 8\text{BF}_3 + 6\text{NH}_3 + 3\text{H}_2\text{O}$
82. $\text{NH}_4\text{BF}_4 \Delta \rightarrow \text{BF}_3 + \text{NH}_3 + \text{HF}$
83. $\text{B}_2\text{O}_3 + 3\text{C} + 3\text{Cl}_2 \Delta \rightarrow 2\text{BCl}_3 + 3\text{CO}$
84. $\text{BF}_3 + \text{NH}_3 \rightarrow \text{F}_3\text{B} \leftarrow \text{NH}_3$
85. $\text{BF}_3 + [\text{C}_2\text{H}_5]_2\text{O} \rightarrow \text{BF}_3 \cdot \text{O}(\text{C}_2\text{H}_5)_2 \text{ (etherate)}$
86. $\text{BF}_3 + \text{F}^- \rightarrow [\text{BF}_4]^-$
87. $\text{BCl}_3 + \text{NH}_3 \rightarrow \text{Cl}_3\text{B} \leftarrow \text{NH}_3$
88. $\text{BCl}_3 + \text{AsH}_3 \text{ selective} \rightarrow \text{Cl}_3\text{B} \leftarrow \text{AsH}_3$
89. $4\text{BF}_3 + 3\text{H}_2\text{O} \rightarrow \text{H}_3\text{BO}_3 + 3\text{HBF}_4$
90. $\text{BCl}_3 + 3\text{H}_2\text{O} \rightarrow \text{H}_3\text{BO}_3 + 3\text{HCl}$
91. $\text{BF}_3 + \text{BCl}_3 \text{ redistribution} \rightarrow \text{BF}_2\text{Cl} + \text{BFCI}_2$
92. $\text{H}_3\text{BO}_3 + 3\text{HF} \rightarrow \text{BF}_3 + 3\text{H}_2\text{O}$
93. $\text{Mg}_3\text{B}_2 + 6\text{H}_3\text{PO}_4 \rightarrow \text{B}_2\text{H}_6 + 3\text{Mg}(\text{H}_2\text{PO}_4)_2$
94. $2\text{BF}_3 + 6\text{NaH} \text{ 450 K} \rightarrow \text{B}_2\text{H}_6 + 6\text{NaF}$
95. $4\text{BF}_3 \cdot \text{OEt}_2 + 3\text{LiAlH}_4 \text{ dry ether} \rightarrow 2\text{B}_2\text{H}_6 + 3\text{LiAlF}_4 + 4\text{OEt}_2$
96. $4\text{BCl}_3 + 3\text{LiAlH}_4 \text{ ether} \rightarrow 2\text{B}_2\text{H}_6 + 3\text{LiCl} + 3\text{AlCl}_3$
97. $2\text{NaBH}_4 + \text{I}_2 \text{ diglyme} \rightarrow \text{B}_2\text{H}_6 + 2\text{NaI} + \text{H}_2$
98. $3\text{NaBH}_4 + 4\text{BF}_3 \cdot \text{OEt}_2 \rightarrow 2\text{B}_2\text{H}_6 + 3\text{NaBF}_4 + 4\text{OEt}_2$
99. $2\text{BCl}_3 + 6\text{H}_2 \text{ electric discharge} \rightarrow \text{B}_2\text{H}_6 + 6\text{HCl}$
100. $\text{B}_2\text{O}_3 + 2\text{Al} + 3\text{H}_2 \Delta \rightarrow \text{B}_2\text{H}_6 + \text{Al}_2\text{O}_3$
101. $\text{B}_2\text{H}_6 + 3\text{O}_2 \rightarrow \text{B}_2\text{O}_3 + 3\text{H}_2\text{O}$
102. $\text{B}_2\text{H}_6 + 6\text{H}_2\text{O} \rightarrow 2\text{H}_3\text{BO}_3 + 6\text{H}_2$
103. $\text{B}_2\text{H}_6 + 6\text{NaOH} \rightarrow 2\text{Na}_3\text{BO}_3 + 6\text{H}_2$
104. $\text{B}_2\text{H}_6 + 6\text{CH}_3\text{OH} \rightarrow 2\text{B}(\text{OCH}_3)_3 + 6\text{H}_2$
105. $\text{B}_2\text{H}_6 + 3\text{Cl}_2 \rightarrow 2\text{BCl}_3 + 6\text{HCl}$
106. $\text{B}_2\text{H}_6 + \text{HCl} \rightarrow \text{B}_2\text{H}_5\text{Cl} + \text{H}_2$
107. $\text{B}_2\text{H}_6 + 2\text{LiH} \rightarrow 2\text{LiBH}_4$
108. $2\text{B}_2\text{H}_6 + 2\text{Na} \rightarrow \text{NaBH}_4 + \text{NaB}_3\text{H}_8$
109. $\text{B}_2\text{H}_6 + 2\text{CO} \Delta, \text{ p} \rightarrow 2\text{BH}_3 \cdot \text{CO} \text{ (symmetrical cleavage)}$
110. $\text{B}_2\text{H}_6 + 2\text{PF}_3 \rightarrow 2\text{BH}_3 \cdot \text{PF}_3$
111. $\text{B}_2\text{H}_6 + 2\text{P}(\text{CH}_3)_3 \rightarrow 2\text{BH}_3 \cdot \text{P}(\text{CH}_3)_3$
112. $\text{B}_2\text{H}_6 + 2\text{N}(\text{CH}_3)_3 \rightarrow 2\text{BH}_3 \cdot \text{N}(\text{CH}_3)_3 \text{ (symmetrical, bulky donor)}$
113. $\text{B}_2\text{H}_6 + 2\text{Et}_2\text{O} \rightarrow 2\text{BH}_3 \cdot \text{OEt}_2$
114. $\text{B}_2\text{H}_6 + 2\text{Et}_2\text{S} \rightarrow 2\text{BH}_3 \cdot \text{SEt}_2$
115. $\text{B}_2\text{H}_6 + 2\text{NH}_3 \text{ low T} \rightarrow [\text{BH}_2(\text{NH}_3)_2]_2 \leftarrow [\text{BH}_4]^- \text{ (unsymmetrical cleavage)}$
116. $3\text{B}_2\text{H}_6 + 6\text{NH}_3 \Delta \rightarrow 2\text{B}_3\text{N}_3\text{H}_6 + 12\text{H}_2$
117. $4\text{NaH} + \text{B}(\text{OCH}_3)_3 \approx 250^\circ\text{C} \rightarrow \text{NaBH}_4 + 3\text{NaOCH}_3$
118. $2\text{MH} + \text{B}_2\text{H}_6 \text{ ether (M = Li, Na)} \rightarrow 2\text{M}[\text{BH}_4]$
119. $\text{NaBH}_4 + \text{LiCl} \rightarrow \text{LiBH}_4 + \text{NaCl}$
120. $\text{BeCl}_2 + 2\text{LiBH}_4 \rightarrow \text{Be}(\text{BH}_4)_2 + 2\text{LiCl}$
121. $3\text{LiBH}_4 + \text{AlCl}_3 \rightarrow \text{Al}(\text{BH}_4)_3 + 3\text{LiCl}$
122. $4\text{LiH} + \text{AlCl}_3 \text{ dry ether} \rightarrow \text{LiAlH}_4 + 3\text{LiCl}$
123. $\text{LiBH}_4 + 2\text{H}_2\text{O} \rightarrow \text{LiBO}_2 + 4\text{H}_2$
124. $3\text{NH}_4\text{Cl} + 3\text{NaBH}_4 \Delta \rightarrow \text{B}_3\text{N}_3\text{H}_6 + 3\text{NaCl} + 9\text{H}_2$
125. $3\text{BCl}_3 + 3\text{NH}_4\text{Cl} \text{ 140 }^\circ\text{C} \rightarrow \text{B}_3\text{N}_3\text{H}_3\text{Cl}_3 + 9\text{HCl}$
126. $\text{B}_3\text{N}_3\text{H}_3\text{Cl}_3 + 3\text{NaBH}_4 \rightarrow \text{B}_3\text{N}_3\text{H}_6 + 3\text{NaCl} + (\text{BH}_3\text{-derived products})$
127. $\text{B}_3\text{N}_3\text{H}_6 + 3\text{HCl} \rightarrow \text{B}_3\text{N}_3\text{H}_9\text{Cl}_3 \text{ (ring addition)}$
128. $\text{B}_3\text{N}_3\text{H}_6 + 9\text{H}_2\text{O} \rightarrow 3\text{H}_3\text{BO}_3 + 3\text{NH}_3 + 3\text{H}_2$
129. $\text{B}_3\text{N}_3\text{H}_6 \text{ strong } \Delta \rightarrow 3\text{BN} + 3\text{H}_2$
130. $\text{B}_2\text{O}_3 + 2\text{NH}_3 \Delta \rightarrow 2\text{BN} + 3\text{H}_2\text{O}$
131. $\text{BCl}_3 + \text{NH}_3 \Delta \rightarrow \text{BN} + 3\text{HCl}$
132. $2\text{B}_2\text{O}_3 + 7\text{C} \approx 2500^\circ\text{C} \rightarrow \text{B}_4\text{C} + 6\text{CO}$
133. $\text{Fe}_2\text{O}_3 + 2\text{Al} \Delta \rightarrow \text{Al}_2\text{O}_3 + 2\text{Fe} + \text{heat}$
134. $\text{Cr}_2\text{O}_3 + 2\text{Al} \Delta \rightarrow \text{Al}_2\text{O}_3 + 2\text{Cr}$
135. $3\text{Mn}_3\text{O}_4 + 8\text{Al} \Delta \rightarrow 4\text{Al}_2\text{O}_3 + 9\text{Mn}$
136. $\text{Al}_2\text{O}_3 + 3\text{C} + 3\text{Cl}_2 \Delta \rightarrow 2\text{AlCl}_3 + 3\text{CO}$
137. $2\text{AlCl}_3(\text{g}) \rightleftharpoons \text{Al}_2\text{Cl}_6 \text{ (dimer; reverse at high T)}$
138. $\text{Al}_2\text{Cl}_6 + 12\text{H}_2\text{O} \rightarrow 2[\text{Al}(\text{H}_2\text{O})_6]^{3+} + 6\text{Cl}^-$
139. $2\text{Al} + 6\text{HCl}(\text{aq}) + 12\text{H}_2\text{O} \rightarrow 2\text{AlCl}_3 \cdot 6\text{H}_2\text{O} + 3\text{H}_2$
140. $\text{Al}(\text{OH})_3 + 3\text{HCl} + 3\text{H}_2\text{O} \rightarrow \text{AlCl}_3 \cdot 6\text{H}_2\text{O}$
141. $2\text{AlCl}_3 \cdot 6\text{H}_2\text{O} \Delta \rightarrow \text{Al}_2\text{O}_3 + 6\text{HCl} + 9\text{H}_2\text{O}$
142. $\text{AlCl}_3 + 6\text{NH}_3 \rightarrow \text{AlCl}_3 \cdot 6\text{NH}_3$
143. $\text{AlCl}_3(\text{aq}) + 3\text{NH}_4\text{OH} \rightarrow \text{Al}(\text{OH})_3 + 3\text{NH}_4\text{Cl}$
144. $\text{Al}(\text{OH})_3 + \text{NaOH} \rightarrow \text{NaAlO}_2 + 2\text{H}_2\text{O}$
145. $[\text{Al}(\text{H}_2\text{O})_6]^{3+} + \text{H}_2\text{O} \rightleftharpoons [\text{Al}(\text{H}_2\text{O})_5\text{OH}]^{2+} + \text{H}_3\text{O}^+$
146. $2\text{Al}(\text{OH})_3 + 3\text{H}_2\text{SO}_4 \rightarrow \text{Al}_2(\text{SO}_4)_3 + 6\text{H}_2\text{O}$
147. $\text{Al}_2(\text{SO}_4)_3 + \text{K}_2\text{SO}_4 + 24\text{H}_2\text{O} \rightarrow \text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$
148. $\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}(\text{s}) \text{ dissolve} \rightarrow \text{K}^+ + \text{Al}^{3+} + 2\text{SO}_4^{2-} + 12\text{H}_2\text{O}$
149. $\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O} \Delta \text{ (burnt alum)} \rightarrow \text{KAl}(\text{SO}_4)_2 + 12\text{H}_2\text{O}$
150. Cathode: $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}(\text{l})$
151. Anode: $\text{C} + 2\text{O}_2 \rightarrow \text{CO}_2 + 4\text{e}^-$
152. Overall: $2\text{Al}_2\text{O}_3 + 3\text{C} \rightarrow 4\text{Al} + 3\text{CO}_2$
153. Bayer: $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{Na}[\text{Al}(\text{OH})_4]$
154. Bayer: $\text{Na}[\text{Al}(\text{OH})_4] \text{ seeding/dilution} \rightarrow \text{Al}(\text{OH})_3 \downarrow + \text{NaOH}$
155. $2\text{Al}(\text{OH})_3 \text{ calcination} \rightarrow \text{Al}_2\text{O}_3 + 3\text{H}_2\text{O}$

BORON FAMILY

SECTION 1 — NCERT CORE

Q1. Count = 13. Only statement 14 is wrong: kernite is $\text{Na}_2\text{B}_4\text{O}_7 \cdot 4\text{H}_2\text{O}$ and borax is $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$, so their degrees of hydration are different. (Statements on the B/Al gradation, the Puga Valley / Sambhar Lake occurrence, the $< 0.0001\%$ crustal abundance, the 10B/11B isotopes with 11B $\sim 80\%$, Al as 3rd-most-abundant element, bauxite and cryolite, and mica in MP/Karnataka/Orissa/Jammu are all NCERT-correct.)

Q2. Count = 14. Only statement 15 is wrong. Filled 3d electrons shield less effectively, not more effectively, than would be required for smooth periodic behaviour; this poor shielding increases effective nuclear attraction and makes Ga slightly smaller than Al. The remaining statements test the causes of the Group 13 anomalies—shell addition, poor d/f shielding, ionisation-enthalpy irregularities, electronegativity behaviour, boron's strong lattice, gallium's low melting point/high boiling point, and the overall density increase—rather than repeating the order list in Q3.

Q3. Count = 18. Wrong: statements 6 and 11. The correct third-ionisation-enthalpy order is $\text{In} < \text{Al} < \text{Tl} < \text{Ga} < \text{B}$, and the correct melting-point order is $\text{Ga} < \text{In} < \text{Tl} < \text{Al} < \text{B}$. The question consolidates the Group 13 orders for atomic/ionic size, IE1-IE3 and ΣIE , electronegativity, density, melting/boiling point, $E^\circ(\text{M}^{3+}/\text{M})$, +1-state stability, metallic/oxide character, and the BF_3 - BI_3 Lewis-acidity, $\text{p}\pi$ - $\text{p}\pi$ -donation and B-X bond-length trends.

Q4. Count = 14. Only 15 is wrong: the inert pair is the ns^2 pair (not np^1); it is the ns^2 electrons that increasingly stay non-bonding down the group, leaving the single np^1 electron to bond in the +1 state.

Q5. Count = 12. Wrong: 13 (borax is alkaline, not acidic — it hydrolyses: $\text{Na}_2\text{B}_4\text{O}_7 + 7\text{H}_2\text{O} \rightarrow 2\text{NaOH} + 4\text{H}_3\text{BO}_3$) and 14 (in $\text{Na}_2[\text{B}_4\text{O}_5(\text{OH})_4] \cdot 8\text{H}_2\text{O}$ four of the 'ten waters' are actually structural $-\text{OH}$ groups, only eight are lattice water).

Q6. Count = 12. Wrong: 11 (the blue cobalt bead is $\text{Co}(\text{BO}_2)_2$, i.e. cobalt(II) metaborate — cobalt is not reduced to the metal) and 12 (B_2O_3 in the bead is a reactive glass-forming acidic oxide that dissolves metal oxides — that is exactly why the bead test works).

Q7. Count = 11. Wrong: 12 (solid H_3BO_3 has planar BO_3 units H-bonded into sheets, not BO_4 tetrahedra), 13 (it is a weak acid, not a strong mineral acid) and 14 (its solubility increases on heating — sparingly soluble cold, much more soluble hot).

Q8. Count = 12. Wrong: 10 (H_3BO_3 is monobasic; it does not lose three protons — it accepts one OH^-) and 14 (it is not a reducing agent). The Lewis-acid equilibrium $\text{B}(\text{OH})_3 + 2\text{H}_2\text{O} \rightleftharpoons [\text{B}(\text{OH})_4]^- + \text{H}_3\text{O}^+$, the $\text{H}_3\text{BO}_3 \rightarrow \text{HBO}_2 \rightarrow \text{B}_2\text{O}_3$ heating and the mild-antiseptic use are all correct.

Q9. Count = 15. Only 15 is wrong: $\text{B}_2\text{O}_3 + \text{H}_2\text{O}$ simply gives back H_3BO_3 — it is not a route to diborane. Statement 16 ($\Delta\text{ch}^\circ \sim -1976 \text{ kJ mol}^{-1}$) and all three preparations ($\text{LiAlH}_4/\text{BF}_3$, NaBH_4/I_2 , NaH/BF_3), the toxicity, spontaneous flammability and complete hydrolysis are NCERT-correct.

Q10. Count = 11. Wrong: 11 (CO does act as a Lewis base: $\text{B}_2\text{H}_6 + 2\text{CO} \rightarrow 2\text{BH}_3 \cdot \text{CO}$), 12 (borazine $\text{B}_3\text{N}_3\text{H}_6$ is inorganic benzene — it contains no carbon) and 13 (ammonia first gives the ionic diammoniate, then borazine on heating, then BN — not BN directly).

Q11. Count = 11. Wrong: 10 (the two bridging H are not equivalent to the four terminal H), 11 (there is no $\text{B}=\text{B}$ bond) and 12 (the two $\text{B}-\text{H}-\text{B}$ bridges are $3\text{c}-2\text{e}$, not ordinary $2\text{c}-2\text{e}$ bonds).

Q12. Count = 11. Wrong: 11 ($[\text{BH}_4]^-$ is tetrahedral, not square planar), 12 (NaBH_4 is a reducing agent / hydride donor) and 13 (borohydrides are made from metal hydride + diborane in ether, $2\text{MH} + \text{B}_2\text{H}_6 \rightarrow 2\text{M}[\text{BH}_4]$, not by hydrolysing B_2H_6).

Q13. Count = 14 — all statements correct. The hardness, high m.p. with low density, very low conductivity, boron fibres in bullet-proof vests and aircraft composites, 10B control rods, borosilicate (Pyrex) glass, glass-wool/fibreglass, flux and glaze use, and medicinal-soap constituent are all in the NCERT text.

Q14. Count = 15. Only statement 15 is wrong: aluminium forms many useful alloys and is not avoided because it cannot alloy. The merged statement 16 is correct: duralumin contains $\text{Cu}/\text{Mn}/\text{Mg}$ and magnalium contains Mg . The remaining physical-property and use statements are retained as correct in the source bank.

Q15. Count = 13. Only 14 is wrong: concentrated HNO_3 makes aluminium passive (it reinforces the oxide film) — it does not attack it. All other reactivity statements (B_2O_3 from amorphous boron, Al_2O_3 film, $\text{EN} \rightarrow$ nitrides, no reaction of B with ordinary acid/alkali, $\text{Al} + \text{dil HCl} \rightarrow \text{H}_2$, $\text{Al} + \text{NaOH} \rightarrow [\text{Al}(\text{OH})_4]^- + \text{H}_2$, EX_3 with TlI_3 the exception, moist-air fuming of AlCl_3) are correct.

Q16. Count = 13. Only pair N is wrong: B_2O_3 is acidic, but the reason given ('boron is the most metallic member') is false — boron is the least metallic. Pairs A–M ($\text{Ga} < \text{Al}$ from 3d screening, ΔH_1 fall $\text{B} \rightarrow \text{Al}$ with size, EN dip then rise, +1 stability toward Tl , TlIII oxidising, Ga thermometry, boron m.p. from its lattice, alkaline borax from hydrolysis, monobasic H_3BO_3 , AlCl_3 moist-air fumes, no $[\text{BF}_6]^{3-}$, $3\text{c}-2\text{e}$ bridges in B_2H_6 , Al passive in conc. HNO_3) are all correctly reasoned.

SECTION 2 — SINGLE-CORRECT OBJECTIVE

Q17. Answer B. Bits 1–4 are the standard cleavage facts: $\text{NH}_3 / 1^\circ / 2^\circ$ amines give unsymmetrical cleavage to $[\text{BH}_2\text{L}_2]^+[\text{BH}_4]^-$; a 3° amine gives the neutral symmetrical adduct $\text{L} \cdot \text{BH}_3$; H- gives $[\text{BH}_4]^-$; CO forms $\text{BH}_3 \cdot \text{CO}$. Bit 5 ('no donor-acceptor interaction') and bit 6 ('every donor gives the same product') are false. So the correct set is 1, 2, 3, 4.

Q18. Answer D. ' TlI_3 ' is $\text{TlI} + (\text{I}_3^-)$ — thallium is +1, not +3, because Tl^{3+} would oxidise I^- to I_2 ($\text{Tl}^{3+}/\text{Tl}^+ E^\circ \approx +1.26 \text{ V} > \text{I}_2/\text{I}^- +0.54 \text{ V}$). I_3^- is linear (sp^3d on the central I) and the two terminal iodines are equivalent but the central one is different. The inert-pair effect stabilises Tl(I) . Correct bits: 1, 2, 3, 6.

Q19. Answer A. Borazine and benzene are isoelectronic, both have planar six-membered rings, borazine has alternating B and N, and its polar $\text{B}\delta^+-\text{N}\delta^-$ bonds make it far more prone to addition and hydrolysis than benzene. Bits 5 and 6 (identical reactivity / identical charge distribution) are false. Correct: 1, 2, 3, 4.

Q20. Answer C. Be–Al diagonal pairs: BeO and Al_2O_3 amphoteric; both chlorides covalent and bridged/associated; Be_2C and Al_4C_3 both give CH_4 on hydrolysis. Be_3N_2 and AlN give NH_3 — statement 4 as printed is fine, but statements 5 ('only strongly basic oxides') and 6 ('ions

unaffected by hydration/hydrolysis') are false, so the complete correct set is 1, 2, 3, 4.

Q21. Answer A. The merged structural audit keeps the non-duplicated borax facts: $\text{Na}_2[\text{B}_4\text{O}_5(\text{OH})_4] \cdot 8\text{H}_2\text{O}$, two trigonal plus two tetrahedral B centres, five B–O–B bridges, four structural OH groups plus eight lattice waters, and one OH group on each B in the standard drawing. The statement assigning four terminal B=O double bonds is false. Correct: 1, 2, 3, 4, 5.

Q22. Answer D. $\text{BF}_3 + \text{LiAlH}_4$ (dry ether) $\rightarrow \text{B}_2\text{H}_6$; diborane is a toxic, electron-deficient gas; complete hydrolysis $\rightarrow \text{H}_3\text{BO}_3 + \text{H}_2$; it is a Lewis acid toward donors; and it has 4 terminal B–H + 2 B–H–B bridges. Statement 5 ('bonding identical to ethane') is false. Correct: 1, 2, 3, 4, 6.

Q23. Answer A. Diborane accepts donors through BH_3 fragments; BF_3 remains a Lewis acid despite $\text{F} \rightarrow \text{B}$ back donation; AlCl_3 forms donor adducts; and BeF_2 can accept F^- to give a fluoro complex. The merged boron-trihalide trend is $\text{BF}_3 < \text{BCl}_3 < \text{BBr}_3 < \text{BI}_3$ in Lewis acidity. Only statement 4 is false. Correct: 1, 2, 3, 5, 6.

Q24. Answer C. From $2\text{BF}_3 + 6\text{NaH} \rightarrow \text{B}_2\text{H}_6 + 6\text{NaF}$, $\text{P} = \text{B}_2\text{H}_6$ and $\text{Q} = \text{NaF}$. Diborane is electron deficient and forms donor adducts. NaF gives a white CaF_2 precipitate with Ca^{2+} . Statement 5 is false because F^- is not oxidised to F_2 by acidified MnO_2 under ordinary laboratory conditions. Correct: 1, 2, 3, 4, 6.

Q25. Answer B. H_3BO_3 has a planar BO_3 part. In water it acts as a weak monobasic Lewis acid and accepts OH^- to form $[\text{B}(\text{OH})_4]^-$. Solid boric acid is hydrogen-bonded. Glycerol or mannitol can form complexes and increase its apparent acidity. Statement 5 is false. Correct: 1, 2, 3, 4, 6.

Q26. Answer D. Borax solution is alkaline because of hydrolysis; $\text{B}(\text{OH})_3$ and $[\text{B}(\text{OH})_4]^-$ are the aqueous species; acidification precipitates boric acid; borax + boric-acid mixtures are used as buffers/primary standards. Statement 3 ('strong mineral acid') and statement 6 ('only B_2O_3 in solution') are false. Correct: 1, 2, 4, 5.

Q27. Answer A. Down Group 13 the +1 state grows more stable (inert-pair effect), so Tl(I) is stabilised and Tl(III) salts are oxidising; ' TlI_3 ' is Tl(I) triiodide. Statement 5 (' Tl(I) less stable than Tl(III) ') and statement 6 ('no Tl(III) compound can exist') are false. Correct: 1, 2, 3, 4.

Q28. Answer C. Alkaline washing-soda solutions attack aluminium; strong alkali forms soluble aluminate + H_2 ; conc. HNO_3 passivates Al; Al is used for transmission conductors (good conductivity, low density); $\text{Al}(\text{OH})_3$ is an antacid. Statement 6 ('keeping water overnight in bare Al utensils is recommended') is not recommended. Correct: 1, 2, 3, 4, 5.

Q29. Answer B. Anhydrous AlCl_3 is covalent, associated as Al_2Cl_6 in vapour/benzene, and 6-coordinate (polymeric) in the solid; moisture $\rightarrow$ hydrolysis and HCl fumes; in water strongly-hydrated $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$ forms. Statement 6 ('hydration enthalpy of Al^{3+} too small to matter') is false. Correct: 1, 2, 3, 4, 5.

Q30. Answer D. Near 373 K $\text{H}_3\text{BO}_3 \rightarrow \text{HBO}_2$; further heating $\rightarrow \text{H}_2\text{B}_4\text{O}_7$ (statement 2 as printed is fine); red heat $\rightarrow \text{B}_2\text{O}_3$. Statement 4 ('borax is the obligatory first dehydration product'), statement 5 (' B_2O_3 easily rehydrated') and statement 6 ('proves boric acid is tribasic') are false. Correct: 1, 2, 3.

Q31. Answer C. $\text{AlN} + 3\text{H}_2\text{O} \rightarrow \text{Al}(\text{OH})_3 + \text{NH}_3$; $\text{Al}_4\text{C}_3 + 12\text{H}_2\text{O} \rightarrow 4\text{Al}(\text{OH})_3 + 3\text{CH}_4$ (a methanide carbide). Statement 5 (' Al_4C_3 gives acetylene') is false — that is CaC_2 . The gases are respectively NH_3 and CH_4 . Correct: 1, 2, 3, 4, 6.

SECTION 3 — ONE-OR-MORE-CORRECT

Q32. Answer A, B, C, D. Bridge B–H contacts (≈ 134 pm) are longer and weaker than terminal B–H (≈ 119 pm); the four terminal bonds are 2c–2e; each B–H–B bridge is 3c–2e; and the 2 B + 4 terminal H are the maximum coplanar set. Bit E (all six H equivalent) and bit F (sp^2 B) are false.

Q33. Answer A, B, C, D, E. $\text{Al}_2(\text{CH}_3)_6$: represented $\text{Al}_2(\text{CH}_3)_6$, two bridging + four terminal methyls, bridging Al–C longer than terminal, electron-deficient 3c–2e multicentre bridges. Bit F (separate Al^{3+} and CH_3^-) is false.

Q34. Answer A, B, C, D, E. $\text{Al}(\text{OH})_3$ is a white gelatinous precipitate, insoluble in excess NH_4OH , dissolves in strong NaOH as aluminate, dissolves in acids, and is used as an antacid. Bit F ('strongly basic, non-amphoteric') is false — it is amphoteric.

Q35. Answer A, B, C, D, E. Chrome-alum chemistry: BaCl_2 precipitates SO_4^{2-} as BaSO_4 ; Na_2CO_3 and $(\text{NH}_4)_2\text{S}$ both give $\text{Cr}(\text{OH})_3$ -type precipitates via hydrolysis; controlled KOH precipitates $\text{Cr}(\text{OH})_3$; excess strong alkali dissolves amphoteric $\text{Cr}(\text{OH})_3$. Bit F (BaCl_2 precipitating chromium as BaCrO_4 from untreated Cr(III)) is false.

Q36. Answer A, B, C, D, E. Colemanite + Na_2CO_3 gives NaBO_2 (A) and borax (B); the balanced flow $\text{Ca}_2\text{B}_6\text{O}_{11} + 2\text{Na}_2\text{CO}_3 \rightarrow 2\text{CaCO}_3 + 2\text{NaBO}_2 + \text{Na}_2\text{B}_4\text{O}_7$ fits; the tetraborate framework has five B–O–B links; strong heating of borax regenerates $\text{NaBO}_2 + \text{B}_2\text{O}_3$. Bit F ('solid NaBO_2 only isolated monomeric BO_2^- ') is false.

Q37. Answer A, B, C, D. Salicylic acid gives a four-coordinate chelated borate; that bis-chelate is optically resolvable; polyhydroxy ligands raise the apparent acidity; and borate + methanol + conc. H_2SO_4 gives volatile $\text{B}(\text{OCH}_3)_3$ that burns green. Bits E ('no reaction, both acids') and F ('boron stays trigonal') are false.

SECTION 4 — MATCH THE COLUMNS

Q38. Answer: A–S, B–Q, C–R, D–P. Complete hydrolysis of diborane gives boric acid, so $\text{W} = \text{H}_3\text{BO}_3$ (S): $\text{B}_2\text{H}_6 + 6\text{H}_2\text{O} \rightarrow 2\text{H}_3\text{BO}_3 + 6\text{H}_2$.

On first heating, H_3BO_3 loses water to give metaboric acid, HBO_2 (Q). Further condensation gives tetraboric acid, $\text{H}_2\text{B}_4\text{O}_7$ (R). Strongest heating finally gives B_2O_3 (P). Therefore A–S, B–Q, C–R, D–P.

Q39. Answer: A–Q; B–P; C–R,S; D–Q. A is a reduction because hydrogen reduces BBr_3 to elemental boron at high temperature. B is the borax-bead test: a copper salt/oxide forms a coloured copper metaborate in the borate glass. C gives white HCl fumes because atmospheric moisture hydrolyses AlCl_3 ; therefore both 'white fumes' and 'hydrolysis' apply. D is aluminothermic reduction: Al reduces Cr_2O_3 to Cr while Al itself is oxidised to Al_2O_3 .

Q40. Answer: A–P, B–Q, C–R, D–S. BF_3 is a trigonal-planar, electron-deficient monomer. Solid anhydrous AlCl_3 has a polymeric/high-coordinate structure. Molecular Al_2Cl_6 contains two bridging chloride atoms. Diborane contains two B–H–B bridges described as three-centre two-electron (3c–2e) bonds.

Q41. Answer: A–P,S; B–Q,S; C–Q,S; D–R. NH_3 , MeNH_2 and Me_2NH are placed in the unsymmetrical-cleavage class and

act as Lewis bases toward electron-deficient boron. Me₃N gives neutral BH₃·NMe₃ by symmetrical cleavage and therefore also represents Lewis-base interaction. CO can likewise donate to BH₃ to form BH₃·CO, so Q and S apply. NaH supplies H[−], converting a BH₃ fragment into BH₄[−]; in the classification used by this question its match is R.

Q42. Answer: A–P, B–Q, C–R, D–S. Borazine, B₃N₃H₆, is called inorganic benzene. Hexagonal BN has a graphite-like layered structure and is often called inorganic graphite. B₄C is an exceptionally hard boron-rich carbide used as an abrasive/armour material. Al₄C₃ is a methanide carbide because hydrolysis gives methane:

$$\text{Al}_4\text{C}_3 + 12\text{H}_2\text{O} \rightarrow 4\text{Al}(\text{OH})_3 + 3\text{CH}_4$$

Q43. Answer: A–P,R,S; B–P,Q,S; C–P,Q,S; D–P,R,S. Borazine shows electron delocalisation in its B₃N₃ ring and hydrolyses readily; boron centres are electron-poor, so the problem set includes Lewis-acid character. Diborane is a Lewis acid, is itself a dimer of BH₃ units, and hydrolyses. Aluminium chloride is a strong Lewis acid, forms the dimer Al₂Cl₆, and hydrolyses in water. BCl₃ is a Lewis acid, has B–Cl π donation/delocalisation in the monomer, and undergoes hydrolysis.

Q44. Answer: Set I: A–S, B–T, C–R, D–P. Set II: A–S, B–R, C–P, D–Q.

Set I: strong heating of borax gives NaBO₂ + B₂O₃ (S); hydrolysis of diborane gives H₃BO₃ (T); excess NH₃ followed by strong heating leads to BN (R); hydrolysis of SiCl₄ followed by heating gives SiO₂ (P).

Set II repeats the first three transformations and adds BCl₃ + LiAlH₄ → B₂H₆, so D–Q.

Q45. Answer: Set I: A–P,Q; B–R,T; C–P,S; D–P,Q. Set II: A–Q; B–R; C–S; D–P.

Set I: borazine hydrolysis gives gaseous H₂ and boric acid, a monobasic acid. Heating borax with NH₄Cl can lead to BN, a graphite-like solid, and boron oxide chemistry, hence R,T in the supplied classification. BF₃ + NaH gives diborane plus NaF; diborane is gaseous and the sodium fluoride product is associated with the paper/brightener description used in the source set. Diborane hydrolysis gives H₂ and H₃BO₃, so P,Q. Set II: heating borax gives NaBO₂ + B₂O₃ (Q); borax + NH₄Cl on heating gives BN (R); acidifying borax gives H₃BO₃ (S); hydrolysis of borax gives boric acid/borate species represented by P.

Q46. Answer: A–Q,R,T; B–P,S,T; C–Q,R,S,T; D–P,R,S,T. B₂H₆ is a covalent gas and on hydrolysis gives boric acid, an oxyacid. H₃BO₃ is a covalent solid and boron is trigonal/sp² in the molecular BO₃ unit. BF₃ is a covalent trigonal-planar gas and hydrolysis gives boron oxyacid species. BI₃ is a covalent solid; monomeric BI₃ has trigonal-planar sp² boron and it hydrolyses to boric acid.

Q47. Answer: Set I: A–Q,S; B–P; C–S; D–Q. Set II: A–R; B–P; C–S; D–T; E–Q.

Set I: hydrated borax contains both trigonal (sp²) and tetrahedral (sp³) boron centres, hence Q,S. Potash alum gives the potassium test with sodium cobaltinitrite, hence P. Borazon means cubic BN; in the supplied classification it is matched to the tetrahedral/sp³ description. Borazole (borazine) has sp² boron, hence Q.

Set II: anhydrous AlCl₃ is a Friedel–Crafts catalyst (R); alum is matched with the antiseptic/use entry (P) in this source set; ultramarine is the complex blue silicate (S); borax is tincal (T); magnalium is an alloy (Q).

SECTION 5 — STATEMENT COUNT

Q48. Count = 10. Correct: borax = Na₂[B₄O₅(OH)₄]·8H₂O; 2 trigonal + 2 tetrahedral B; five B–O–B links; zero B=O π bonds; strong heat → NaBO₂ + B₂O₃; the glassy bead dissolves metal oxides; CoO → blue bead; borax is a primary standard in acidimetry; the solution is alkaline from hydrolysis; borax → H₃BO₃ → B₂O₃ → B is a real route. Wrong: ‘all four B tetrahedral’ and ‘contains a B=B π bond’.

Q49. Count = 10. Correct: 12 valence electrons; four terminal 2c–2e; two bridge 3c–2e; bridge B–H longer and weaker than terminal; the 2 B + 4 terminal H are the maximum coplanar set; bridge H above/below that plane; B is sp³; 3° amines give symmetric BH₃·L; NH₃/MeNH₂/Me₂NH are grouped as unsymmetrical-cleavage donors. Wrong: ‘every B–H is an ordinary 2c–2e bond’ and ‘bridge bonds are shorter and stronger’.

Q50. Count = 10. Correct: anhydrous AlCl₃ covalent; moist-air partial hydrolysis with HCl fumes; large hydration enthalpy stabilises ionic Al species in water; [Al(H₂O)₆]³⁺ octahedral; [Al(OH)₄][−] tetrahedral; Al₂Cl₆ dimer in vapour; solid AlCl₃ polymeric/6-coordinate; chloride bridges reduce electron deficiency; Al can show CN > 4; anhydrous and hydrated forms need not share a bonding description. Wrong: ‘AlCl₃·6H₂O = dry monomeric AlCl₃’ and ‘hydrated Al³⁺ is destabilised by hydration’.

Q51. Count = 10. Correct: B₃N₃H₆; ‘inorganic benzene’; alternating B and N; sp² B and N in the planar ring; B–N back donation in the classical picture; more reactive than benzene toward addition/hydrolysis; hydrolysis → H₃BO₃ + NH₃; H₂ in the source representation; high-T NH₃/borane chemistry gives BN; h-BN is graphite-like. Wrong: ‘identical reactivity toward HCl’ and ‘every B–N bond non-polar’.

Q52. Count = 10. Correct: weak monobasic Lewis acid; accepts OH[−] rather than donating three protons; hydrogen-bonded solid; heating → HBO₂ → B₂O₃; formed by hydrolysis of BCl₃ and of B₂H₆; from acidification of borax; polyhydroxy compounds raise apparent acidity; salicylic acid gives a four-coordinate chelate; borate + methanol + conc. H₂SO₄ → green-edged flame. Wrong: ‘strong tribasic protonic acid’ and ‘no reaction with salicylic acid because both are acids’.

Q53. Count = 11. Correct: borax → H₃BO₃ → B₂O₃; B₂O₃ + Mg → B; crude B needs acid washing; B dissolves in hot conc. H₂SO₄; conc. HNO₃ passivates Al; Al used for transmission conductors; Al + NaOH mixtures evolve H₂ (drain cleaners); washing-soda solutions attack Al as aluminate; Al(OH)₃ as antacid; AlN → NH₃; Al₄C₃ → CH₄. Wrong: ‘calcium is the preferred clean reducing metal’ (magnesium is used).

Q54. Count = 3. Statements 1, 2 and 4 are correct in the supplied source set; statements 3, 5 and 6 are not counted as correct there.

SECTION 6 — NUMERICAL RESPONSE

Q55. Answer = 6. In B₂H₆, the two B atoms and four terminal H atoms are coplanar; the two bridging H atoms lie above and below this plane. Therefore the maximum coplanar atom count is 6.

Q56. Answer = 4. Diborane contains four terminal B–H bonds, and each terminal B–H bond is an ordinary 2c–2e bond.

Q57. Answer = 2. Diborane has two B–H–B bridges. Each bridge is described as one 3c–2e bond, so the number is 2.

Q58. Answer = 12. Total valence electrons in $\text{B}_2\text{H}_6 = 2(3) + 6(1) = 12$.

Q59. Answer = 2. The $[\text{B}_4\text{O}_5(\text{OH})_4]^{2-}$ ion contains two trigonal BO_3 -type boron centres and two tetrahedral BO_4 -type centres; hence $\text{sp}^2 \text{ B} = 2$.

Q60. Answer = 2. In $[\text{B}_4\text{O}_5(\text{OH})_4]^{2-}$, two of the four boron centres are tetrahedral, so $\text{sp}^3 \text{ B} = 2$.

Q61. Answer = 5. The standard tetraborate framework contains five bridging oxygen atoms linking boron centres, i.e. five $\text{B}-\text{O}-\text{B}$ bridges.

Q62. Answer = 14. Five bridging O atoms contribute 10 $\text{B}-\text{O}$ σ bonds (two per bridge), and four terminal $-\text{OH}$ groups contribute 4 more $\text{B}-\text{O}$ σ bonds. Total = $10 + 4 = 14$.

Q63. Answer = 8. Structural borax is $\text{Na}_2[\text{B}_4\text{O}_5(\text{OH})_4] \cdot 8\text{H}_2\text{O}$, so it contains eight lattice-water molecules per formula unit.

Q64. Answer = 6. Borazine has a planar six-membered B_3N_3 ring. All three B and all three N ring atoms are sp^2 -hybridised, giving 6 sp^2 ring atoms.

Q65. Answer = 12. In the standard planar borazine molecule, the six ring atoms and all six directly attached H atoms lie in the molecular plane. Maximum coplanar atoms = 12.

Q66. Answer = 1.5 mol. From $\text{B}_2\text{H}_6 + 6\text{H}_2\text{O} \rightarrow 2\text{H}_3\text{BO}_3 + 6\text{H}_2$, one mole B_2H_6 gives six moles H_2 . Thus $0.25 \times 6 = 1.5$ mol H_2 .

Q67. Answer = 6 mol. From $\text{B}_2\text{H}_6 + 6\text{CH}_3\text{OH} \rightarrow 2\text{B}(\text{OCH}_3)_3 + 6\text{H}_2$, one mole B_2H_6 gives two moles $\text{B}(\text{OCH}_3)_3$. Therefore 3.0 mol B_2H_6 gives 6.0 mol ester.

Q68. Answer = 2 mol. $\text{Na}_2\text{B}_4\text{O}_7 + 7\text{H}_2\text{O} \rightarrow 2\text{NaOH} + 4\text{H}_3\text{BO}_3$. Therefore 0.50 mol tetraborate gives $0.50 \times 4 = 2.0$ mol H_3BO_3 .

Q69. Answer = 2 mol. Strong dehydration follows $2\text{H}_3\text{BO}_3 \rightarrow \text{B}_2\text{O}_3 + 3\text{H}_2\text{O}$. Hence 4.0 mol H_3BO_3 gives 2.0 mol B_2O_3 .

Q70. Answer = 3 mol. From $2\text{Al} + 2\text{NaOH} + 6\text{H}_2\text{O} \rightarrow 2\text{Na}[\text{Al}(\text{OH})_4] + 3\text{H}_2$, 2.0 mol Al gives 3.0 mol H_2 when alkali and water are in excess.

Q71. Answer = 6 mol. $2\text{NaBH}_4 + \text{I}_2 \rightarrow 2\text{NaI} + \text{B}_2\text{H}_6 + \text{H}_2$. Each 2 mol NaBH_4 gives 2 mol gaseous products (1 mol $\text{B}_2\text{H}_6 + 1$ mol H_2). Therefore 6 mol NaBH_4 gives 6 mol total gas.

Q72. Answer = 6. Correct are (i), (ii), (iii), (iv), (v) and (vii). Statements (vi) and (viii) are false because $\text{p}\pi-\text{p}\pi$ overlap is strongest for $\text{B}-\text{F}$, while Lewis acidity increases from BF_3 to BI_3 .

Q73. Answer = 3. The formula of beryl is $\text{Be}_3\text{Al}_2\text{Si}_6\text{O}_{18}$; therefore $n = 3$.

Q74. Answer = 2. Al_2Cl_6 contains two bridging chloride atoms and four terminal chloride atoms. Therefore the number of bridging chlorides is 2.

SECTION 7 — INTEGRATED QUESTIONS

Q75. Answer C. $\text{NH}_3/\text{MeNH}_2/\text{Me}_2\text{NH} \rightarrow$ unsymmetrical cleavage; 3° amines $\rightarrow$ symmetric $\text{BH}_3 \cdot \text{L}$; $\text{NaH} \rightarrow [\text{BH}_4]^-$; bridge $\text{B}-\text{H}$ longer/weaker than terminal. CO is not an amine-type reagent (bit 5) and $1^\circ/2^\circ/3^\circ$ amines do not give identical products (bit 6). Correct set 1, 2, 3, 4.

Q76. Answer B. Sodium peroxoborate carries active oxygen; in alkaline medium it furnishes an H_2O_2 -type oxidant that oxidises $\text{Cr}(\text{III}) \rightarrow$ chromate (represented as CrO_4^{2-} in alkali). It is not a reducing agent (bit 5) and its chemistry is peroxide-based (bit 6). Correct set 1, 2, 3, 4.

Q77. Answer D. Hydrolysis of BF_3 gives HF, which attacks the SiO_2 of glass; boric acid is the boron-containing product; so moisture changes the practical compatibility of BF_3 with glassware. BCl_3 hydrolysis gives HCl, not HF (bit 5), and silica is not inert to aqueous HF (bit 6). Correct set 1, 2, 3, 4.

Q78. Answer A. Borax-bead colours: $\text{Co} \rightarrow$ blue, $\text{Cr} \rightarrow$ green, $\text{Mn} \rightarrow$ violet/amethyst, $\text{Cu} \rightarrow$ blue/green. Ca^{2+} gives no diagnostic transition-metal colour (bit 5 false). The colours arise from metal borate/metaborate species in the glass. Correct set 1, 2, 3, 4, 6.

Q79. Answer C. Borax-acid $\rightarrow \text{H}_3\text{BO}_3 \cdot \Delta \rightarrow \text{B}_2\text{O}_3$; $\text{B}_2\text{O}_3 + 3\text{Mg} \rightarrow 2\text{B} + 3\text{MgO}$; acid wash removes Mg species; high-T $\text{BBr}_3 + \text{H}_2 \rightarrow$ purer boron. 'Direct reduction of borax with water' (bit 5) is false. Correct set 1, 2, 3, 4, 6.

Q80. Answer A, B, C, D. Hall-Héroult: Al_2O_3 is dissolved in molten cryolite (+ CaF_2); the bath lowers the working temperature and raises conductivity; carbon/graphite electrodes are used and the anode carbon is consumed ($\text{C} + 2\text{O}_2 \rightarrow \text{CO}_2 + 4\text{e}^-$); aluminium is deposited at the cathode. Bits E (aqueous AlCl_3 electrolysis) and F (Al at the anode) are false.

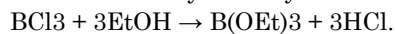
Q81. Answer 5. Of the six practical-aluminium statements, five are correct: conc. HNO_3 can be stored in Al (passivation); Al wire in transmission lines; $\text{Al} + \text{NaOH}$ in drain cleaners; washing soda attacks Al as soluble aluminate; $\text{Al}(\text{OH})_3$ as antacid. 'Keeping water overnight in bare Al utensils is recommended' is not — count = 5.

Q82. Answer 7. Diagonal-relationship statements true: $\text{BeO}/\text{Al}_2\text{O}_3$ amphoteric; $\text{Be}_2\text{C}/\text{Al}_4\text{C}_3 \rightarrow \text{CH}_4$; Be/Al nitrides $\rightarrow \text{NH}_3$; Be/Al chlorides bridged/associated; $\text{B}_2\text{O}_3/\text{SiO}_2$ acidic; $\text{BCl}_3/\text{SiCl}_4$ covalent and hydrolysable; B/Si form hard refractory carbide/boride-type solids. The eighth ('identical formulae for every hydride and halide') is false $\rightarrow 7$.

Q83. Answer 13. Of the fifteen master-audit statements, thirteen are correct; the two false ones are 'every Group 13 halide is a monomeric trigonal molecule in every state' (AlCl_3 dimerises; TlX are ionic) and 'all six hydrogens in B_2H_6 are equivalent $2\text{c}-2\text{e}$ terminal H' (two are $3\text{c}-2\text{e}$ bridge H).

Q84. Answer: (a) D. (b) 0.

(a) A is correct because BF_3 accepts F^- to form tetrahedral $[\text{BF}_4]^-$. B represents hydrolysis of BF_3 through a hydroxyfluoroborate species and is accepted in the source set. C is ordinary alcoholysis of BCl_3 :



D is incorrect as written: BCl_3 normally forms a 1:1 Lewis adduct with pyridine; the written neutral 1:2 species $\text{Cl}_3\text{B}(\text{py})_2$ would require five-coordinate boron and is not the standard product.

(b) None of A–D is scientifically correct in the form stated. A is false because BF_3 hydrolysis is not simply "complete" under all conditions; fluoride/ HBF_4 species can remain. B is false because anhydrous AlCl_3 is predominantly covalent. C is false in the comparison intended here; AlCl_3 is not simply a weaker Lewis acid than BCl_3 . D is false because each Al in molecular Al_2Cl_6 has coordination number 4 (two terminal Cl and two bridging Cl), not 6. Hence count = 0.

Q85. Answer:

(a) $\text{A} = [\text{BH}_2(\text{NH}_3)_2]^+[\text{BH}_4]^-$, $\text{B} = \text{B}_3\text{N}_3\text{H}_6$ (borazine), $\text{C} = \text{H}_2$. The element in C is hydrogen, so its atomic number is 1. The cationic part $[\text{BH}_2(\text{NH}_3)_2]^+$ contains 2 NH_3 molecules.

At low temperature ammonia causes unsymmetrical cleavage of diborane:



On heating the ammonia–borane system eliminates hydrogen and forms borazine.

(b) Lewis acids = 3: H^+ , B_2H_6 and CH_3^+ . H^+ and CH_3^+ accept electron pairs directly; B_2H_6 accepts donors through electron-deficient BH_3 fragments.

Lewis bases = 9: benzene, H_2O , CH_3^- , $(\text{C}_2\text{H}_5)_2\text{O}$, PH_3 , acetone, H^- , CO and NO . Each can donate an electron pair (or, for benzene, π electron density) in the classification used by the question. Therefore acids = 3 and bases = 9.

SECTION 8 — EXTENDED STATEMENT COUNT

Q86. Count = 9. Correct: $\text{NaBO}_3 \cdot 4\text{H}_2\text{O}$ shorthand; dimeric anion $\text{Na}_2[(\text{HO})_2\text{B}(\text{O}-\text{O})_2\text{B}(\text{OH})_2] \cdot 6\text{H}_2\text{O}$; two genuine $\text{O}-\text{O}$ peroxo bridges; both B tetrahedral; prepared by electrolysis of borate or $\text{NaBO}_2 + \text{H}_2\text{O}_2$; warming above $\sim 80^\circ\text{C}$ releases H_2O_2 ; used as an oxygen bleach; oxidises $\text{Cr}(\text{III}) \rightarrow$ chromate; a convenient solid source of active oxygen. Wrong: ‘the “peroxo” is only nominal’, ‘ NaBO_2 contains the same $\text{O}-\text{O}$ linkage’, ‘acts only as a reducing agent’.

Q87. Count = 9. Correct: BF_3 green flame ($\text{CaF}_2 + \text{conc. H}_2\text{SO}_4$ on the borate); $\text{B}(\text{OCH}_3)_3$ green flame ($\text{CH}_3\text{OH} + \text{conc. H}_2\text{SO}_4$); conc. H_2SO_4 removes water; turmeric/curcumin $\rightarrow$ red then greenish-black with NH_3 ; the green colour is a covalent volatile species not an ion; both tests need boron first converted to boric acid; moist BF_3 attacks glass via HF ; a large amount of sodium masks the flame. Wrong: ‘boron gives a crimson atomic flame like Li ’, ‘dry BF_3 etches silica’, ‘ $\text{B}(\text{OCH}_3)_3$ is non-volatile’.

Q88. Count = 10. Statements 1–10 are correct; 11 and 12 are false. Borazine is a planar $\text{B}_3\text{N}_3\text{H}_6$ ring with alternating B and N, isoelectronic with benzene, but the $\text{B}\delta^+-\text{N}\delta^-$ bonds make it less aromatic and much more reactive than benzene. It readily undergoes addition. Addition of 3HCl gives $\text{B}_3\text{N}_3\text{H}_9\text{Cl}_3$, and hydrogenation gives $\text{B}_3\text{N}_3\text{H}_{12}$. Hydrolysis gives boric acid, ammonia and hydrogen in the source representation. Borazine can be obtained from the diborane–ammonia system; B-trichloroborazine is formed from $\text{BCl}_3/\text{NH}_4\text{Cl}$ and can be reduced to borazine. Statement 11 is false because borazine is not more aromatically stable than benzene; statement 12 is false because borazine is far more easily hydrolysed.

Q89. Count = 10. Correct statements are 1–9 and 12; statements 10 and 11 are false. “ TlI_3 ” is $\text{Tl}^+(\text{I}_3^-)$, not $\text{Tl}(\text{III})$ iodide; I_3^- is linear. “ GaCl_2 ” and “ InCl_2 ” are mixed-valence salts written $\text{Ga}^+[\text{GaCl}_4]^-$ and $\text{In}^+[\text{InCl}_4]^-$. $\text{Tl}(\text{I})$ shows useful similarities to $\text{K}(\text{I})$ and $\text{Ag}(\text{I})$; Tl_2S is black and thallium compounds are cumulative poisons. The +1 oxidation state becomes more stable down Group 13. Statement 10 is false because GaCl_2 does not contain a genuine paramagnetic $\text{Ga}(\text{II})$ centre. Statement 11 is false because TlI_3 is not $\text{Tl}^{3+} + 3\text{I}^-$. B_2Cl_4 contains a genuine $\text{B}-\text{B}$ bond and is non-eclipsed in the gas phase.

Q90. Count = 9. Correct statements are 1–9; the final three are false. Anodising makes aluminium the anode in dilute H_2SO_4 and thickens the oxide film. AlCl_3 and BF_3 are Lewis-acid Friedel–Crafts catalysts; AlCl_3 can be tied up as $[\text{AlCl}_4]^-$ during the reaction. HBF_4 is a strong acid containing tetrahedral $[\text{BF}_4]^-$, while KBF_4 is sparingly soluble. ^{10}B is valuable for neutron absorption; B_4C is a hard armour/abrasive material; GaAs and GaN are important semiconductors; covalent $\text{Al}(\text{BH}_4)_3$ and $\text{Be}(\text{BH}_4)_2$ contain bridging hydride interactions. The anodising statement using Al as cathode in molten cryolite is false; neutral AlCl_3 is not simply recovered unchanged in Friedel–

Crafts acylation; thermite works because formation of Al_2O_3 is highly favourable, not because its $\Delta_f\text{H}$ is less negative.